



Adaptive human-computer interaction for industry 5.0: A novel concept, with comprehensive review and empirical validation

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ABSTRACT

This paper explores the domain of adaptive Human-Computer Interaction (HCI) within the emerging context of Industry 5.0, which marks the transition from Industry 4.0 by emphasizing human-centric approaches and collaboration between humans and intelligent systems. It focuses on enhancing user experience while maintaining high security standards. The complexity of intelligent industrial environments introduces various challenges, including vulnerabilities caused by faults that can propagate across multiple layers and heterogeneous systems. Predictive maintenance becomes more complicated due to the need for advanced monitoring across interconnected systems, and the dynamic nature of these environments demands seamless adaptation. This is essential for fostering harmony between human operators and intelligent systems in real-time. However, the increasing complexity of these environments reveals the limitations of the current HCI models, making adaptability a basic requirement. This paper carries a complete 141-paper review regarding research on HCI along with a detailed exposition of the core architecture, enabling technologies, and real-world application examples. Furthermore, it presents an innovative conceptual model for implementing an HCI system that dynamically adapts to environmental changes, user behavior, diverse user profiles, and varying accessibility needs, specifically within the context of fault detection and diagnosis systems. The proposed approach has been rigorously validated through empirical studies, demonstrating its effectiveness and practical applicability. To improve resiliency and efficiency in such a smart industrial system as is vulnerable, the interface would adapt itself to user behavior and diversity. It addresses the challenges that arise during this dynamic and complex environment that would ensure a secure and seamless interaction.

1. Introduction

The transition from Industry 4.0 (I4.0) to Industry 5.0 (I5.0) marks an important shift in people's roles as actors in the industrial systems. While I4.0 was about how Cyber-Physical Systems (CPS) could create self-sufficient systems and productivity gains, this new I5.0 is all about human-centric systems. It incorporates all human aspects of creativity and decision-making into the processes while achieving human well-being in production processes, creating equivalence between

computing and human insight through HCI. Fig. 1 shows how, in such a new scenario, I5.0 creates an environment for integration of the productivity, efficiency and precision of computers with human creativity and worker well-being. This would set the stage for true harmony between intelligent systems and human operators, so that technology does not only improve operations but also can work along with human skills and intuitions.

I5.0 is human-centric, emphasizing the security of individuals and their interactions with advanced technologies (Alves, Lima, and Gaspar,

Abbreviations: AES-256,, Advanced Encryption Standard 256-bit; AHCI,, Adaptive Human-Computer Interaction; AI,, Artificial Intelligence; AR,, Augmented Reality; AWS,, Amazon Web Services; BBCA,, Behavioral Biometrics Continuous Authentication; CCPA,, California Consumer Privacy Act; CPS,, Cyber-Physical Systems; FDD,, Fault Detection and Diagnosis; GDPR,, General Data Protection Regulation; HAR,, Human Activity Recognition; HCI,, Human-Computer Interaction; HRV,, Heart Rate Variability; I4.0,, Industry 4.0; I5.0,, Industry 5.0; IoT,, Internet of Things; IVE,, Intelligent Virtual Environments; LoRaWAN,, Long Range Wide Area Network; LPWANS,, Low-Power Wide-Area Networks; MFA,, Multi-Factor Authentication; ML,, Machine Learning; MQTT,, Message Queuing Telemetry Transport; NDN,, Named Data Networking; NLP,, Natural Language Processing; QoE,, Quality of Experience; SM,, Smart Manufacturing; UX,, User Experience; VR,, Virtual Reality; XAI,, Explainable Artificial Intelligence.

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2023). An adaptive and responsive HCI is critical in this context, particularly for handling system failures, preventive maintenance, and the dynamic complexity of the industrial environment. Such a system ensures operational efficiency while maintaining a safe, efficient work environment for human operators, facilitating smooth interactions between humans and technology. For this reason, Fault Detection and Diagnosis (FDD) and predictive maintenance is one of the most relevant specific cases of I5.0 since it corresponds directly to the consideration of human-centricity and focuses on worker well-being and safety

Recent methodologies have focused on improving HCI in I5.0, with an emphasis on scalability, personalization, and real-time adaptation. Rani, Jining, Shoukat, Shoukat and Nawaz (2024) suggested a three-level HCI model (product, economic, and ecological) aimed at product personalization and better machine collaboration. This approach highlighted the importance of user-centric designs for improving system effectiveness. Yang, Liu and Morgan (2024) discussed challenges related to data quality, particularly the impact of noisy or insufficient data on predictive maintenance and decision-making accuracy. They also identified the rising cognitive load on workers, which may hinder decision-making in highly automated environments. Wang, Zhou, Li, Yang, Zheng, Song and Wang (2024) introduced the human digital twin framework to improve human-robot collaboration and personalized interaction by integrating human traits into I5.0 systems (Zhu and Zhu, 2022). Yang, Liu and Morgan (2024) also proposed real-time personalization based on dynamic human attributes, allowing continuous system adaptation to operator needs. Peruzzini, Prati and Pellicciari (2024) introduced an augmented digital twin, which facilitates real-time data sharing and adaptive human-machine interfaces to enhance human-automation collaboration. Finally, Martini, Bellisario and Coletti (2024) emphasized the integration of human-centered Artificial Intelligence (AI) for additive manufacturing, optimizing design, and improving system performance.

While these methodologies contribute valuable insights, they also present certain limitations that need to be addressed. Rani, Jining, Shoukat, Shoukat and Nawaz (2024) explored HCI in additive manufacturing but did not provide solutions for scalability in complex environments. It's necessary to enhance dynamic task assignments and collaborative dashboards to support real-time decision-making and improve scalability, particularly in multifaceted production scenarios. Yang, Liu and Morgan (2024) discussed data quality challenges but did

not offer an effective method for reducing data overload in real-time especially in high-stress situations, leading to decision fatigue. It highlighted the increasing mental load on workers as automation grows, but it lacked an effective strategy to manage this cognitive overload. To address this, it's essential to focus on cognitive accessibility and workflow customization, ensuring that only relevant data is shown to operators, which reduces cognitive overload and improves decision-making accuracy.

Wang, Zhou, Li, Yang, Zheng, Song and Wang (2024) introduced the human digital twin framework but faced difficulties in scaling it for real-world applications. It's needed to integrate personalized workflows and adaptive training modules, ensuring this framework can be applied across diverse environments. Similarly, Yang, Liu and Morgan (2024) proposed real-time personalization but did not address the challenge of continuously updating data for accurate personalization. It's crucial to incorporate real-time adaptability, ensuring that the system can adjust dynamically to changes in operator profiles and cognitive states. Peruzzini, Prati and Pellicciari (2024) provided a theoretical framework but lacked practical solutions for integrating human factors into complex industrial settings. It's essential to improve integration with real-time behavioral insights and collaborative dashboards, enhancing communication and task distribution among operators and machines. Finally, Martini, Bellisario and Coletti (2024) noted difficulties in AI integration, particularly with data scarcity and the complex interactions between AI, materials, and human operators. It's needed to prioritize cross-functional collaboration and dynamic task assignment, making AI systems more accessible and actionable for operators, overcoming the complexity of these interactions.

To address these key limitations in existing systems, it's necessary to develop a new conceptual model that ensures a more inclusive and efficient HCI framework. Additionally, by addressing user security, diversity, and well-being, it provides a robust solution for real-world implementation in I5.0.

- This paper aims to advance the state of Adaptive HCI (AHCI) systems by, (1) conducting a **comprehensive review and assessment of recent studies** and technologies in AHCI, focusing on their adaptivity, security, User Experience (UX), and application to industrial contexts, (2) offering an **in-depth technology assessment** of key HCI technologies, evaluating their strengths and limitations across

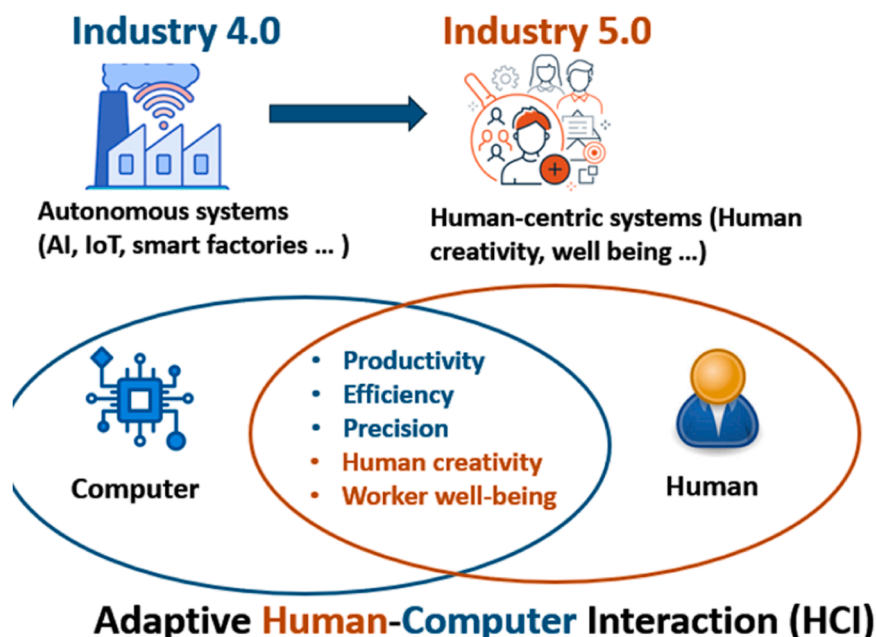


Fig. 1. The role of HCI in the transition from I4.0 to I5.0.

multiple criteria such as adaptivity, security, UX, cost, complexity, portability, privacy concerns, and energy consumption, (3) focusing on **key technologies** highlighting three standout technologies: Behavioral biometrics, intelligent agent-based HCI, and edge-computing for real-time adaptive security in HCI and provides a detailed analysis of their evolution and applications, and finally (4) proposing the **implementation and empirical validation of a new conceptual model** that addresses existing limitations in AHCI frameworks. This model emphasizes, specifically, the case of FDD and advanced maintenance or even new concepts of self-healing to automatically enhance human security within I5.0. By incorporating personalized workflows, cognitive and physical accessibility and real-time adaptability, it aims to streamline decision-making, minimize cognitive overload, and ensure a safe, efficient, and inclusive human-machine interaction. These improvements not only enhance operational reliability but also create a dynamic environment where human well-being and advanced technological processes coexist harmoniously.

This paper is organized as presented in Fig. 2: 2 describes a cursory study of what HCI is, from definitions to the evolution of research to highlights of recent review papers on HCI. It also outlines the method adopted to do a full state-of-the-art review that will serve as a basis for subsequent analysis. In 3, we provide an elaborate review and evaluation of adaptive HCI technologies. Findings from this research will unlock the strengths and limitations of the current approaches, paving the way for proposing our novel conceptual model. As it is needed to use technologies to elaborate the proposed model, this in-depth state of the art and technology assessment is necessary for an effective design of the proposed conceptual model.

After collecting the information needed from the in-depth review of AHCI technologies, 4 introduces the proposed model with a discussion of methodological steps, features, and potential for real-world applications. This section also offers empirical validation of the model, demonstrating its effectiveness and results obtained from its application. To reflect on the broader implications of our work, 5 addresses the problems related to the real-world application of the method proposed, future research directions and challenges as well as the broader picture in adaptive HCI. Finally, 6 concludes with a summary of key findings and restatement of the importance of adaptive HCI for the user well-being in I5.0. Such a structured approach guarantees thorough exploration of the subject and emphasizes the necessity for our contributions.

2. Concept and evolution of research in HCI

2.1. Concept and characteristics of HCI

AHCI systems are designed to dynamically adjust their behavior, interface, and functionality based on user needs, preferences, and contexts (Emond, 2021; Ahmad, Mubin, and Orlando, 2017). These systems aim to enhance UX and user interface efficiency by monitoring and analyzing user interactions, often incorporating user modeling, personalization, and computational learning techniques (Payne and Howes, 2022; Fairclough, 2022). AHCI extends to various domains, including neuroadaptive technology that utilizes neurophysiological data (Fairclough, 2022), context-aware systems that respond to environmental factors and adaptive games for cognitive evaluation (Kandemir and Kose, 2022). The core principle involves creating interfaces capable of modifying their structure and behavior in response to changing user contexts, thereby providing universal usability and addressing the increasing complexity of computer systems (Stigberg, 2020). This approach to HCI aims to improve usability, user satisfaction, and overall interaction quality by tailoring the digital experience to individual users and specific usage situations (Emond, 2021; Ahmad, Mubin, and Orlando, 2017). Adaptive user interfaces monitor user interactions, analyze usage patterns, and present personalized interfaces based on accumulated data, including adapting to users' emotions, personality, and memory of past interactions (Ahmad, Mubin, and Orlando, 2017). The field continues to evolve, addressing challenges in universal usability through the plasticity of interfaces and the integration of advanced technologies for more intuitive and responsive HCIs (Payne and Howes, 2022; Fairclough, 2022; Kandemir and Kose, 2022; Stigberg, 2020).

AHCI transforms user interaction by adapting systems to individual needs and contexts, enhancing usability and satisfaction. Building on that, the next sub-section proposes an in-depth review on advancements in AHCI, focusing on methodologies and key research that address adaptivity and security challenges in complex, interconnected environments.

2.2. Evolution of research in adaptive and secure HCI

2.2.1. Review methodology

To provide a comprehensive review of adaptive and secure HCI, as illustrated in Fig. 3, the selected studies were published between 2018 and 2024 with primary keywords such as "human-computer

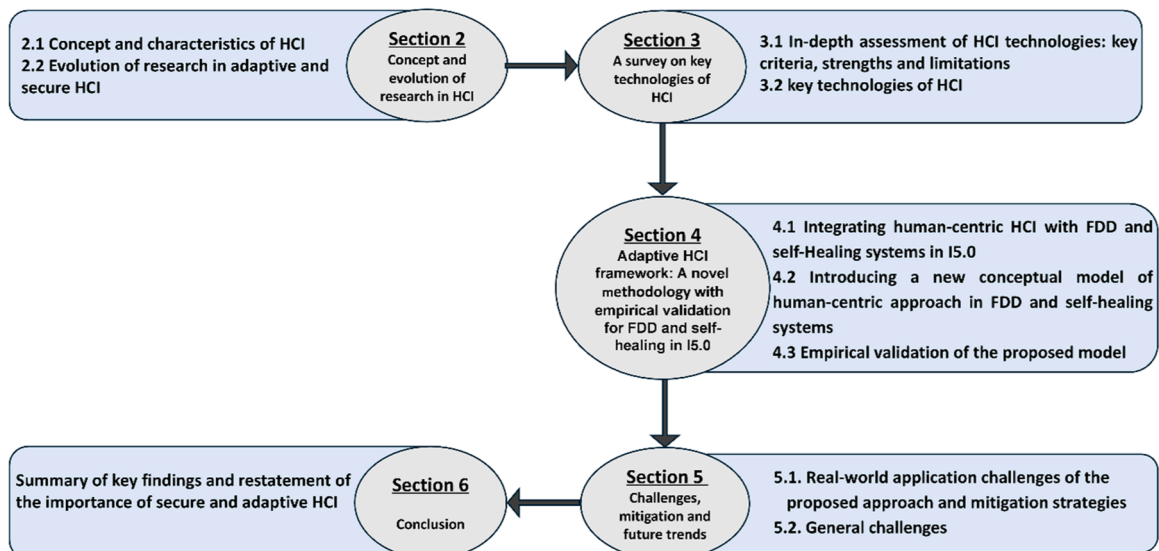


Fig. 2. The overall structure of the paper.

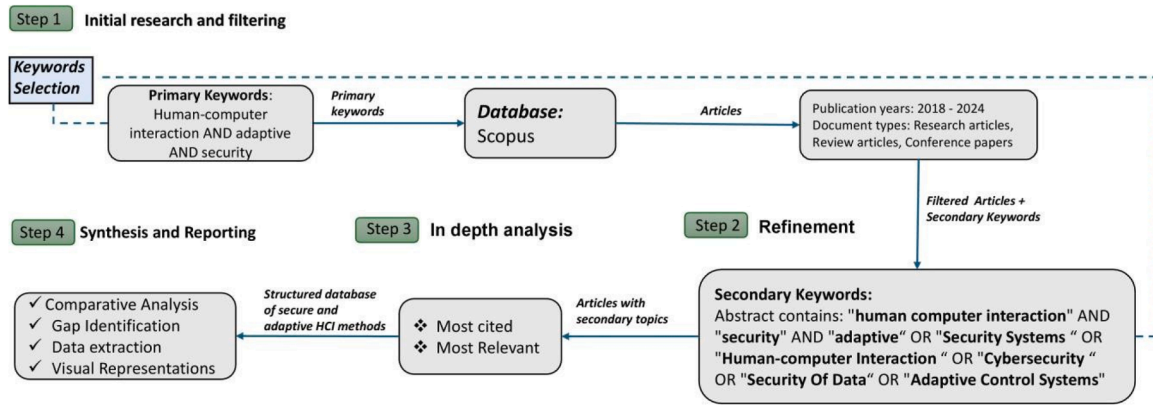


Fig. 3. Review methodology.

Table 1

Summary of the important surveys related to HCI.

Ref.	HCI concepts and benefits	HCI enabling technologies	HCI applications	HCI Novel Methodology	HCI challenges	HCI research directions	Remarks
Bian et al., (2022)	+	++	++	+-	+-	+-	Focuses on the novel use of human body capacitance in activity recognition.
J. Liu, (2024)	++	+-	++	-	++	++	Provides an overview of ChatGPT's technical foundations, HCI impact, and applications.
Wang et al., (2021)	++	++	+-	-	++	++	Discusses HCI in construction hazard recognition, covering technologies and future directions.
Alsamhi et al., (2024)	++	-	+-	-	++	++	Explores integration of human-centered AI and CPS for cognitive situation awareness.
Hauptman, Flathmann, and McNeese, (2024)	++	+-	+-	-	+-	++	Reviews human factors in adaptive autonomy, covering concepts, technologies, and challenges.
Darejeh et al., (2024)	++	+-	+-	+-	++	++	Offers a framework for selecting cognitive load measurement methods in interface usability.
Paneru et al., (2024)	++	+-	++	-	+-	++	Reviews UI and UX design concepts and evolving trends.
Ravuri, (2024)	++	+-	+-	-	++	++	Reviews Human Activity Recognition (HAR) techniques, datasets, and applications.
Reddy, (2024)	++	++	+-	-	++	++	Focuses on key HCI techniques for explainable AI (XAI) systems.
Burtscher and Gerling, (2024)	++	-	+-	NA	++	++	Analyzes HCI research on neurodivergence in the workplace.
Park and McKilligan, (2018)	++	+-	+-	-	+-	++	Reviews HCI and design thinking integration, identifying commonalities and differences.
Pagar et al., (2024)	++	+-	+-	+-	++	++	Analyzes HCI within smart living environments, discussing key concepts, enabling technologies, and applications.
Mourtzis et al., (2023)	++	+-	+-	+-	++	+-	Explores the human-centric metaverse in I5.0, discussing applications and challenges.
Gupta et al., (2023)	++	+-	+-	-	++	++	Surveys HCI and natural habits-based behavioral biometric modalities for user recognition schemes.
Obaigbena et al., (2024)	+-	+-	+-	NA	++	++	Reviews AI-driven human-robot interaction, covering enabling technologies and applications.
Mishra, Satpathy, and Pati, (2023)	++	+-	+-	-	++	++	Reviews HCI applications in healthcare, focusing on XAI for patient safety.
Zhao et al., (2023)	+-	-	+-	+-	++	++	Develops a distributed adaptive secure consensus control strategy using edge computing.
This paper	++	++	++	++	++	++	Reviews key enabling technologies for adaptive HCI and suggests a new conceptual model.

Notes: ++ ⇒ High coverage; +- ⇒ Medium coverage; - ⇒ Low coverage; NA ⇒ Not applicable.

interaction", "adaptive" and "security" in the Scopus database. Using the "AND" operator, we refined our search to include secondary keywords like "Security Systems", "Cybersecurity" and "Security of Data" resulting in a focused set of relevant articles. From these, we selected the most cited and most relevant papers, totaling a substantial collection of key research. These papers were then analyzed to synthesize insights on architecture, enabling technologies, and application scenarios of HCI, providing a well-rounded understanding of the current state of the field.

Table 1 systematically organizes and summarizes key surveys in HCI, assigning a depth level to various components such as **concepts**, **enabling technologies**, **applications**, **methodologies**, **challenges**, and **research directions**. This structured categorization enables a comparative view of how different studies have contributed to the evolution of the field. It offers a clear quantitative measure of each study's focus, making it easier to identify gaps, trends, and standout contributions.

The results in the table reveal that many papers excel in concepts, applications, and research directions, with notable contributions from studies such as Wang et al., (2021) and Reddy, (2024). However, there is a significant gap in novel methodology, where most studies either lack contributions (0) or offer limited innovation (1), such as Bian et al., (2022) and Mourtzis et al. (2023). In contrast, this paper stands out with a full score (2) across all components, particularly in novel methodology, where it not only proposes a new conceptual model but also validates it through empirical studies, bridging the gap between theory and practical application.

The radar chart in Fig. 4 specifically highlights the **HCI novel methodology** dimension as presented in Table 1, showcasing the comparative performance of the papers in this critical area. Most studies demonstrate limited coverage, with scores of 0 or 1, reflecting a reliance on existing frameworks or partial innovation. For instance, J. Liu, (2024) and Wang et al., (2021) scored 0, indicating no contribution to methodology development, while Bian et al., (2022) and Mourtzis et al., (2023) scored 1, showing moderate contributions without extensive validation.

In contrast, this paper achieves a full score 2, standing out for introducing a novel methodology and validating it empirically. This approach not only strengthens its theoretical foundation but also demonstrates its practical viability, setting it apart as a comprehensive and impactful contribution to the field. The radar chart effectively emphasizes this strength, underscoring the paper's unique role in advancing HCI through both innovation and validation.

2.2.2. A statistical overview of global research on HCI

The analysis of Fig. 5 reveals several key trends in the research landscape related to HCI secure and adaptive systems. Fig. 5(a) shows that China, the United States, and India lead in the number of publications, indicating strong research output in these countries. Fig. 5(b) highlights that lecture notes in computer science are the predominant source for these publications, followed by other specialized journals and conference proceedings, suggesting a concentrated dissemination through well-established academic channels. Fig. 5(c) indicates that most of the research is concentrated on computer science, with significant contributions from mathematics and engineering, underscoring the interdisciplinary nature of this field (Ding et al., 2024). Finally, Fig. 5(d) illustrates the fluctuations in research activity over time, with notable peaks in 2020 and 2022, potentially reflecting increased interest or developments in the field during these periods. Overall, these figures collectively underscore the global and interdisciplinary nature of research in this area, with particular emphasis on key contributing countries, sources, and subjects.

3. A survey on key technologies of HCI

3.1. In-depth assessment of HCI technologies: key criteria, strengths and limitations

While much of the existing literature on HCI has been developed in the context of I4.0, with a strong emphasis on automation and efficiency, there is a growing recognition of the need to revisit these technologies within the framework of I5.0. For instance, adaptive user interfaces that were once designed primarily for efficiency must now be reimagined to enhance HCI fostering environments where technology supports rather than supplants human capabilities (Haoqi Wang et al., 2023). Therefore, HCI technologies are essential for enhancing the way humans interact with digital systems across various domains.

3.2. Evaluating HCI technologies: Key criteria and visualization of key insights

Table 2 offers a detailed evaluation of these technologies based on several critical criteria, including adaptivity, security, UX, cost, complexity, portability, privacy concerns, and energy consumption. Each technology is rated on a scale from 0 to 3, where 0 indicates no relevance and 3 represents high relevance or impact. The heatmap

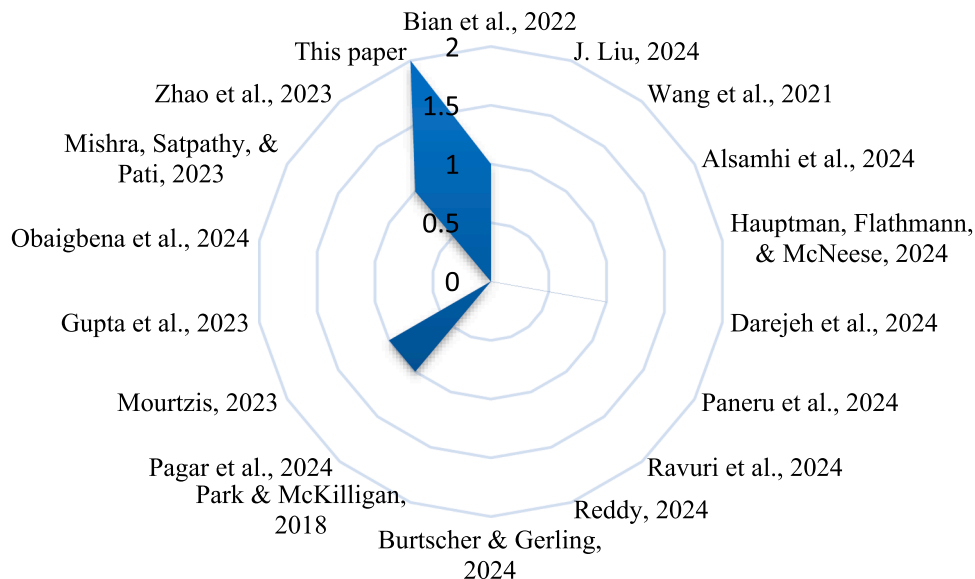


Fig. 4. Comparison of contributions to novel methodology in HCI surveys.

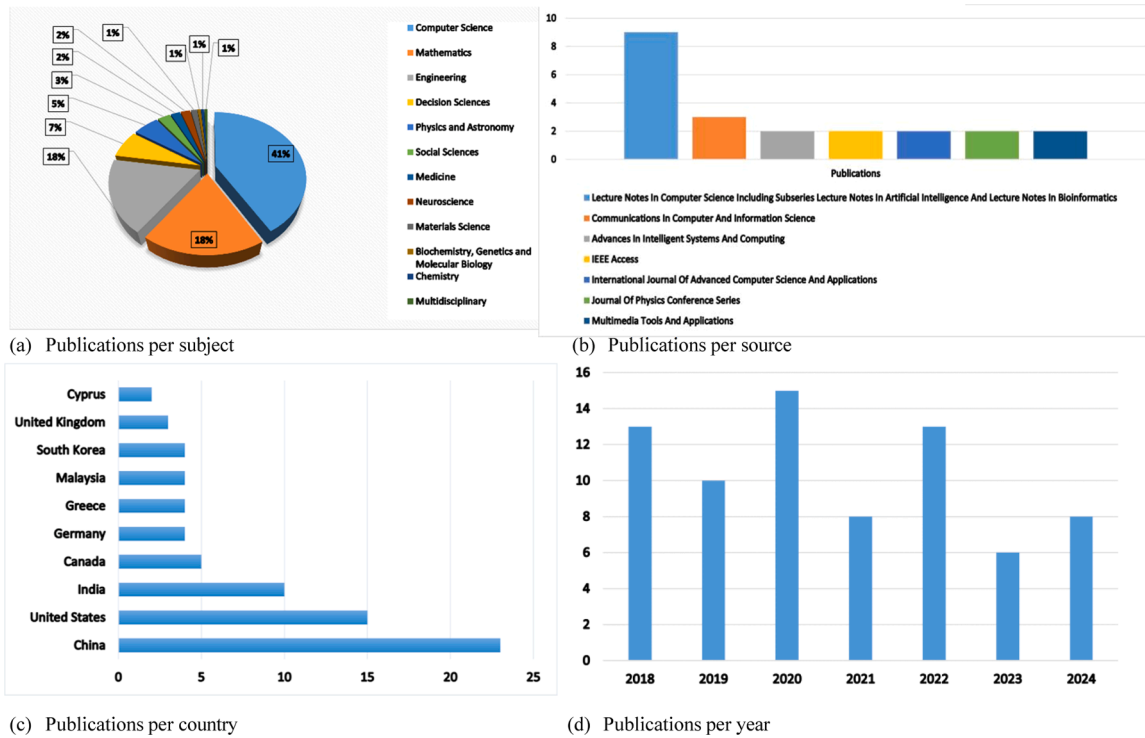


Fig. 5. Research evolution of HCI in the Scopus database from 2018 to 2024.

visualization in the table clarifies the relevance and impact of each criteria across the different technologies. For example, the adaptivity criteria evaluates how well a technology can adapt to changing conditions or user needs, with scores ranging from 0 (not relevant) to 3 (high relevance). Technologies with higher scores in adaptivity, such as conversational agents, demonstrate their flexibility in dynamic environments.

Similarly, the stacked bar provides quick visual cues about which technologies are most energy-intensive, cost-effective, or pose significant privacy concerns, aiding stakeholders in making informed decisions about technology adoption and implementation. Additionally, the technologies are categorized by their domains of application, offering insights into their practical utility in different contexts.

Strengths of advanced HCI technologies

The analysis reveals that most HCI technologies excel in adaptivity and UX, effectively meeting user needs and enhancing interaction quality. Fig. 6 illustrates the core strengths of advanced technologies emphasizing the most outstanding technology for each criteria. Tools like conversational interfaces (Landay, Oliver, and Song, 2019), VR and AR (Riegler et al., 2019; S.-C. Chen, 2021) stand out for their ability to create immersive and adaptable experiences. Tools like interactive visualizations (Sadeghi Milani et al., 2024) and natural language explanations (Raikov, Giretti, Pirani, Spalazzi, & Guo, 2024) further enhance data analysis and decision support. Portability emerges as a significant strength in technologies like wearable sensors for HAR (Ferrara, 2024) and intelligent agent-based HCI (Collins, Koehler, and Lynch, 2024), making them suitable for diverse environments, including healthcare and fitness.

Additionally, behavioral biometrics (Kaji and Aoyama, 2022), intelligent agent-based HCI, and edge-computing are highlighted for their exceptional performance across multiple criteria, including adaptivity, security, UX, cost, complexity, portability, privacy concerns, and energy consumption. These technologies demonstrate superior versatility, providing benchmarks for advancing HCI technologies. Their

adaptability and robustness reflect the growing emphasis on human-centric design, aligning closely with IS.0 principles.

Limitations of HCI technologies

Despite their strengths, HCI technologies face significant challenges, particularly in areas such as security, cost, complexity, privacy, and energy consumption. Fig. 7 provides detailed visualization, highlighting the specific technologies that encounter difficulties in each criteria. Security concerns are most prominent in technologies like reinforcement learning from human feedback (H. Lee and Park, 2024), quantum HCI (Germanakos, 2024), and behavioral biometrics (Cheng Wang et al., 2024), which are crucial for cybersecurity and fraud detection but require substantial resources. In contrast, simpler tools like prototyping (Popoola, Adama, Okeke, and Akinoso, 2024) and usability evaluation methods (Carvalho et al., 2024) focus less on security, prioritizing accessibility and lower complexity. Cost and complexity remain barriers for resource-intensive technologies such as quantum HCI (Germanakos, 2024) and deep learning for HAR (Kumar, Chauhan, and Awasthi, 2024), limiting their widespread adoption. Privacy concerns are another critical issue for technologies handling personal data, such as conversational interfaces (Landay, Oliver, and Song, 2019) and behavioral biometrics (Cheng Wang et al., 2024), raising ethical and regulatory challenges. Moreover, energy consumption varies widely (Luzolo, Elrawashdeh, Galland, Outay, and Gomes, 2023): for example, AR/VR (Riegler et al., 2019; S.-C. Chen, 2021) are less energy-efficient compared to more sustainable options like wearable sensors (Ferrara, 2024) and prototyping tools. The analysis also highlights trade-offs between privacy and UX, where technologies like conversational interfaces and emotional expression in HCI (Votintseva, Johnson, and Villa, 2024; Esposito, Esposito, and Vogel, 2015) offer rich UX but raise significant privacy concerns. Similarly, the importance of portability is underscored by technologies like inertial measurement units and wearable motion sensing prototypes, which drive adoption in industries like healthcare and fitness but introduce additional design and implementation challenges.

To conclude, a holistic evaluation of these technologies highlights their strengths and limitations, offering insights for stakeholders to make informed decisions regarding their adoption. The exemplary performance of behavioral biometrics, intelligent agent-based HCI, and edge-computing underscores their potential to set benchmarks in HCI research and applications. By addressing challenges such as security, cost, privacy, and energy consumption, these technologies can be optimized for scalable, efficient, and user-friendly solutions in advanced HCI systems. Future research must critically evaluate gaps in these technologies to ensure their continued evolution and broader adoption.

Through this state of the art, several key challenges have emerged that are critical for advancing the field:

Challenge 1: Achieving secure and adaptive HCI through continuous authentication methods that evolve with user behaviors and environments.

Challenge 2: Developing intelligent systems for more adaptive and responsive HCI solutions that cater to diverse user needs and contexts.

Challenge 3: Integrating advanced computing paradigms to enhance real-time adaptivity and security in HCI systems while maintaining optimal performance.

These challenges reflect the interdisciplinary nature of HCI research and the growing need for innovative solutions that can keep pace with technological advancements and user expectations.

3.3. Key technologies of HCI

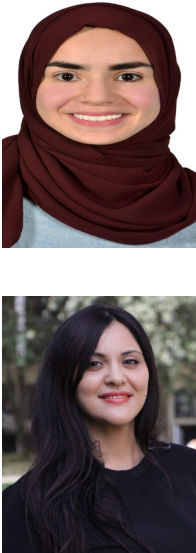
3.3.1. Behavioral biometrics in adaptive HCI (Challenge 1)

This section addresses challenge 1: Achieving secure and adaptive HCI through continuous authentication methods that evolve with user behaviors and environments. According to publication search results from 2018 to 2024 regarding the topic of behavioral biometrics and HCI, in the Scopus database, the number of publications shows a rapid growth trend in recent years. Fig. 8 showcases some of the most insightful contributions like the BehaveSense framework (Y. Yang et al., 2019), Behavioral Biometrics Continuous Authentication (BBCA) (Skalkos, Stylios, Karyda, and Kokolakis, 2021) and the framework B2auth (Smith

and Doe, 2024). The exploration of BBCA technologies highlights various methods and their effectiveness in enhancing user security on digital platforms (Stylios and Aegean, 2023). One significant study (Skalkos, Stylios, Karyda, and Kokolakis, 2021) focuses on user privacy attitudes towards BBCA technologies, integrating privacy concerns, trust in technology, and innovativeness using protection motivation theory. This study, conducted through a cross-sectional survey of 778 smart-phone users, reveals that privacy concerns significantly impact the intention to use BBCA technology (Buriro, Gupta, Yautsiukhin, and Crispo, 2021). It also identifies innovativeness as a crucial factor influencing usage intention, emphasizing the importance of reliability and trustworthiness in security technologies.

Dynamic biometric and behavioral authentication methods, as illustrated in Fig. 9, are divided into two main categories: touch-based methods and body-based methods. **Touch-based methods** include continuous authentication approaches such as dynamic touchscreen biometrics (Leyfer and Spivak, 2019), which analyze touchscreen input events using advanced classifiers like gradient boosting to distinguish legitimate users from intruders with high accuracy. Similarly, touch gesture authentication frameworks (Alariki and Manaf, 2014) continuously monitor unique user touch patterns to enhance security. Gesture-based recognition methods, like passive touch biometrics (H. Xu, Zhou, & Lyu, 2014), provide seamless verification during interactions, delivering contextual fine-grained analysis to ensure reliability. **Body-based methods** include capacitive scanning techniques, exemplified by the bodyprint method (Holz, Buthpitiya, and Knaust, 2015), which utilizes capacitive touchscreens to scan body parts such as ears, palms, and fingers, achieving high identification accuracy and improving mobile device security. Additionally, EEG and hand gesture-based systems, such as CORMORANT (Hintze and Schmidt, 2019), integrate multiple biometric modalities for adaptive and risk-aware authentication, while IoT applications combine EEG signals with gestures (Elshenaway and Guirguis, 2021) for effective verification. In VR, solutions like RubikBiom (Mathis, Fawaz, and Khamis, 2020) merge behavioral biometrics with knowledge-based inputs to achieve high classification accuracy.

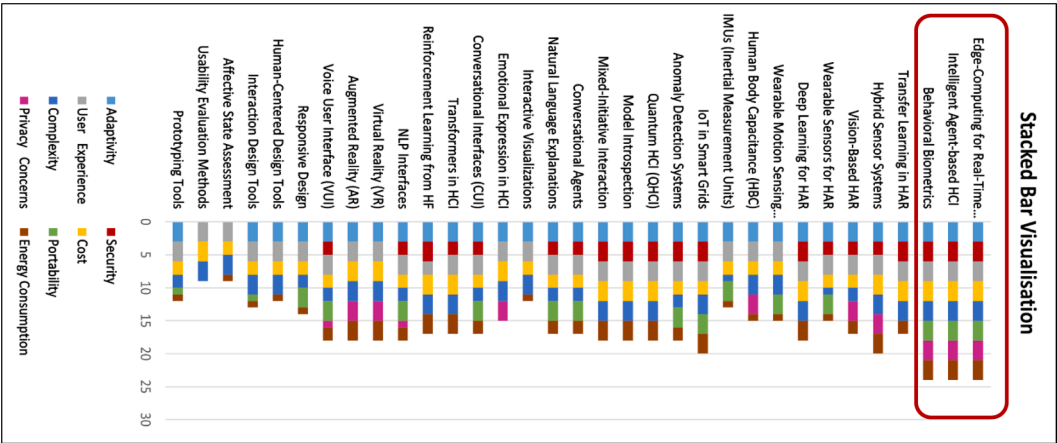
Table 2
Heatmap and stacked bar visualization of HCI technologies based on most relevant criteria.



(continued on next page)

Table 2 (continued)

Technology name	Adaptivity	Security	UX	Cost	Complexity	Portability	Privacy concerns	Energy consumption	Domain of application
Prototyping tools	3	0	3	2	2	1	0	1	Software development, product design, UI design
Usability evaluation methods	0	0	3	3	3	0	0	0	UI design, web development, interactive systems
Affective state assessment	0	0	3	2	3	0	0	1	Gaming, educational software, emotional applications
Interaction design tools	3	0	3	2	3	1	0	1	Web and mobile design, digital products
Human-centered design tools	3	0	3	2	3	0	0	1	Consumer electronics, healthcare, public systems
Responsive design	3	0	3	2	2	3	0	1	Cross-platform Web and mobile applications
Voice user interface	3	2	3	2	2	3	1	2	Smart home devices, virtual assistants
Augmented Reality (AR)	3	0	3	3	3	0	3	3	Gaming, education, healthcare, marketing
Virtual reality (VR)	3	0	3	3	3	0	3	3	Gaming, education, smart environments, simulations
Natural Language Processing (NLP) interfaces	3	2	3	2	2	3	1	2	Customer service, virtual assistants, education
Reinforcement learning	3	3	2	3	3	0	0	3	Chatbots, customer service, personalized learning
Transformers in HCI	3	2	3	3	3	0	0	3	NLP applications, translation, content generation
Conversational user interfaces	3	2	3	2	2	3	0	2	Smart speakers, Internet of Things (IoT) devices, virtual agents
Emotional expression in HCI	3	0	3	3	3	0	3	0	Mental health, customer service, education
Interactive visualizations	3	0	3	2	3	0	0	1	Data analysis, dashboards, decision support
Natural language explanations	3	2	3	2	2	3	0	2	Customer service, legal AI, medical diagnostics
Conversational agents	3	2	3	2	2	3	0	2	Customer support, healthcare, AI systems
Mixed-initiative interaction	3	3	3	3	3	0	0	3	Medical decision-making, legal analysis
Model introspection	3	3	3	3	3	0	0	3	Research, regulatory, high-stakes decisions
Quantum HCI	3	3	3	3	3	0	0	3	Cybersecurity, finance, national security
Anomaly detection systems	3	3	3	2	2	3	0	2	Smart grids, cybersecurity, IoT
IoT in smart grids	3	3	3	2	3	3	0	3	Energy management, healthcare, smart cities
Inertial measurement units	3	0	3	2	1	3	0	1	Fitness tracking, healthcare, gaming
Human body capacitance	3	0		2	3	0	3	1	Fitness tracking, collaborative work
Wearable motion sensing prototypes	3	0	3	2	3	3	0	1	Fitness, healthcare, collaboration
Deep learning for HAR	3	3	3	3	3	0	0	3	Healthcare, smart homes, security
Wearable sensors for HAR	3	2	3	2	1	3	0	1	Healthcare, fitness, rehabilitation
Vision-based HAR	3	2	3	2	2	0	3	2	Surveillance, sports analysis, HCI
Hybrid sensor systems	3	2	3	3	3	0	3	3	Healthcare monitoring, smart environments, industry
Transfer learning in HAR	3	3	3	3	3	0	0	2	Healthcare, smart environments, security
Behavioral biometrics	3	3	3	3	3	3	3	3	Cybersecurity, fraud detection, access control
Intelligent agent-based HCI	3	3	3	3	3	3	3	3	Interactive systems, personalized learning, automation
Edge-computing for real-time adaptivity in HCI	3	3	3	3	3	3	3	3	IoT, smart cities, industrial control



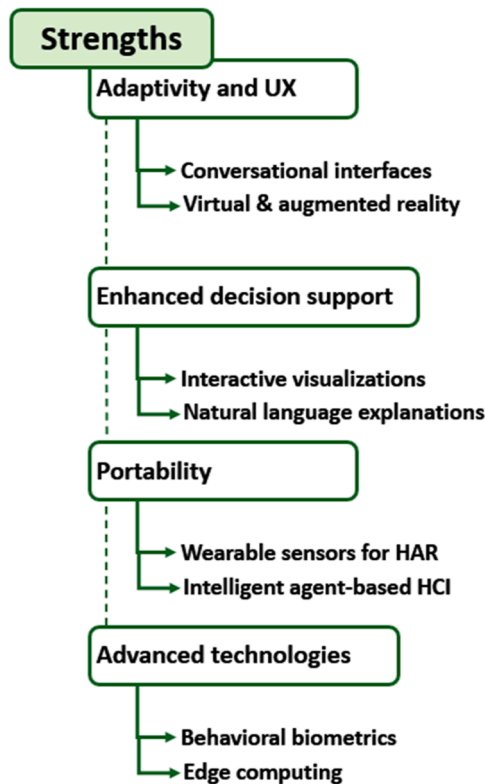


Fig. 6. Core strengths of advanced HCI technologies across key criteria.

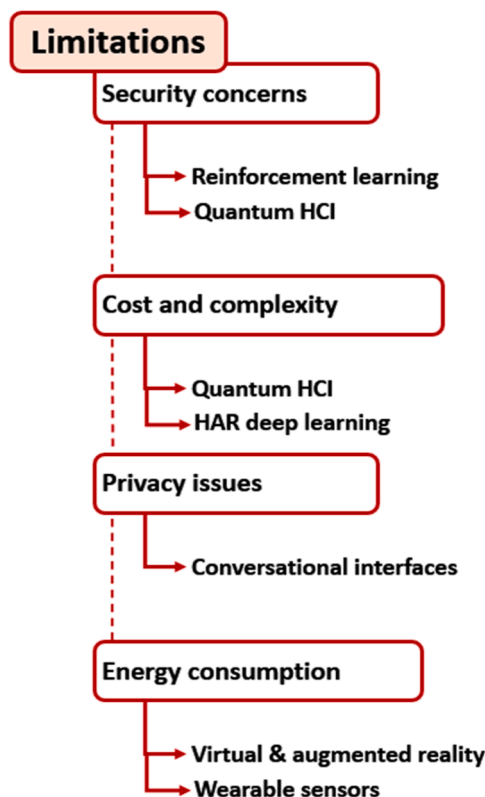


Fig. 7. Limitations of advanced HCI technologies across key criteria.

Touch-based methods are characterized by seamless user verification, high accuracy, and adaptability to user behavior, while body-based methods excel in multimodal authentication (Thomas and Preetha

Mathew, 2023), providing enhanced security for diverse environments such as IoT and VR systems (Sin and Munteanu, 2020). This categorization reflects the strengths and features of dynamic biometric authentication approaches, as shown in Fig. 9, offering a structured understanding of their applications and potential.

Emerging biometric modalities, such as iris recognition, gait recognition, and behavioral biometrics, offer new opportunities for enhancing authentication systems (Tiwari et al., 2024; Wang and Liu, 2020). However, research on privacy-protecting techniques for behavioral data (Hanisch et al., 2021) highlights an imbalance in focus: While traits like voice receive significant attention, others, such as eye-gaze and brain-waves, remain largely overlooked.

Innovative authentication methods continue to emerge. An agent-based intelligent HCI system in mixed reality (Alqahtani et al., 2019) uses adaptive features to improve user interaction (CarreraRivera, RegueraBakhache, Larrinaga, and Lasa, 2023) and information security. Similarly, an adaptive continuous authentication method based on keystroke and mouse usage patterns (Chen et al., 2019; Gałazkiewicz and Wójtowicz, 2024) demonstrates high accuracy with low false acceptance and rejection rates. A recent user identification approach using real-world HCI behavior data, including keyboard and mouse inputs (Khan et al., 2024), also achieves promising results. Security-focused advancements further strengthen authentication reliability. A scalable behavioral dynamics authentication model (Idowu et al., 2023) incorporates keystroke typing patterns with encrypted behavioral data, ensuring both accuracy and security. Likewise, a practical behavioral biometrics model for login authentication (Solano et al., 2020) delivers reasonable false acceptance and rejection rates, even with minimal training data. Finally, a real-world authentication method using keystroke and mouse dynamics (Shi et al., 2023) shows strong performance.

While these advancements highlight the growing potential of behavioral biometrics for security and UX across digital platforms, they also raise concerns about privacy, security, and potential misuse. The next paragraph explores strategies to mitigate these risks and ensure the ethical implementation of continuous, non-intrusive authentication in adaptive HCI systems.

Fig. 10 illustrates the three primary components of a continuous authentication security framework: privacy protection, security measures, and compliance & monitoring. These elements work together to ensure the ethical and secure use of continuous authentication technologies (Bharadwaj and Gupta, 2021), mitigating risks and maintaining compliance with data protection regulations. Continuous and non-intrusive authentication systems offer significant security benefits but also introduce challenges related to privacy, security, and regulatory compliance. To address these concerns, various mitigation strategies are implemented:

- Privacy protection: Techniques such as differential privacy, data minimization, and federated learning help limit exposure of sensitive user data while preserving authentication accuracy (Johnson and White, 2021).
- Security measures: Encryption safeguards data during transmission and storage, Multi-Factor Authentication (MFA) adds an extra layer of security, and data isolation restricts unauthorized access (Anitha et al., n.d.).

Compliance & monitoring: Adhering to privacy-by-design principles ensures alignment with regulations like General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA), while audit trails track data access for accountability. Additionally, anomaly detection powered by AI helps identify suspicious behaviors, preventing potential misuse.

A well-informed consent process and transparent data usage policies further enhance trust in these systems. By integrating these security measures, continuous authentication can be both effective and ethical,

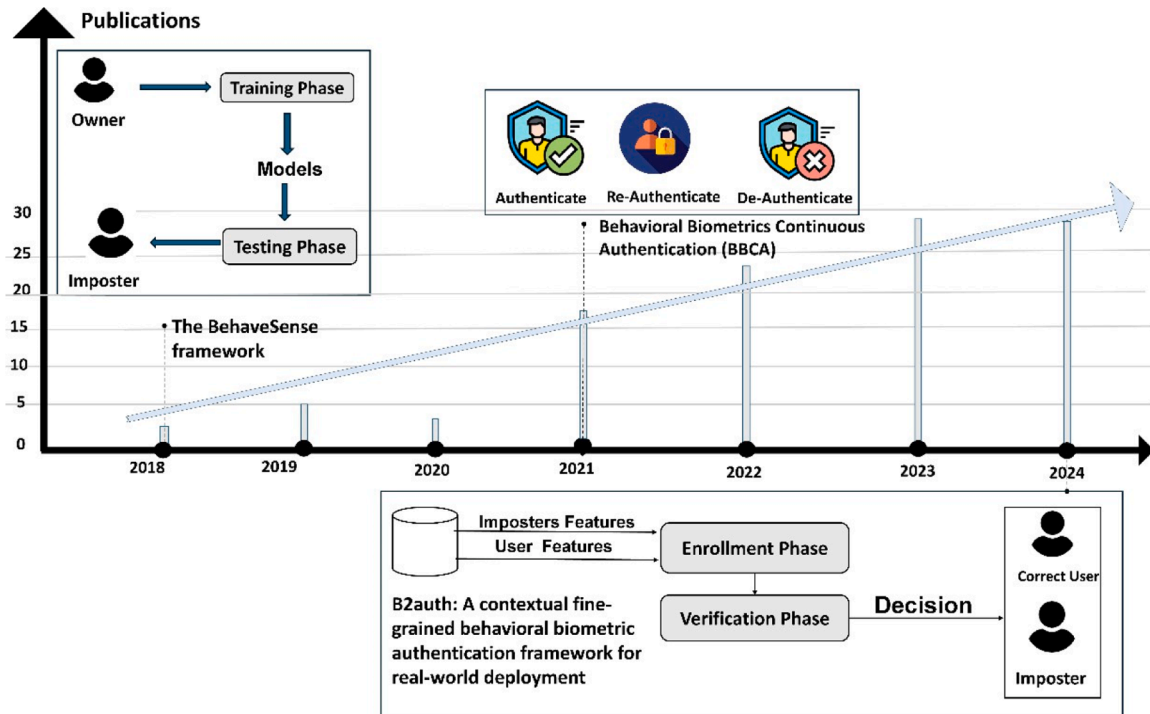


Fig. 8. Evolution of behavioral biometrics in adaptive HCI.

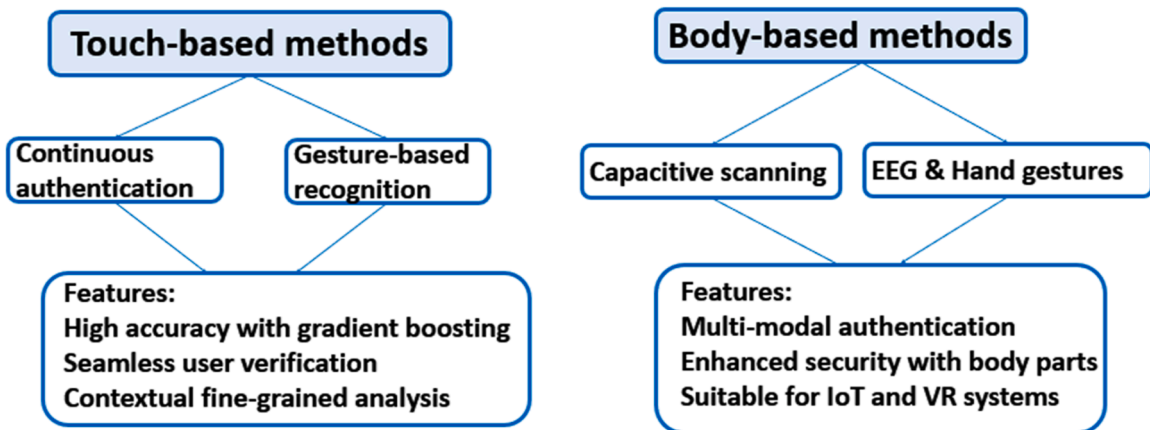


Fig. 9. Dynamic biometric authentication methods: touch-based and body-based approaches.

supporting secure interactions in adaptive HCI environments.

3.3.2. Intelligent agent-based HCI Systems (Challenge 2)

The field of agent-based systems for HCI is rapidly evolving, encompassing diverse applications and research areas. These systems aim to provide efficient, intuitive interfaces while addressing challenges like user privacy, environmental adaptability, and robust security. A key focus is on developing intelligent systems that cater to diverse user needs and contexts, encapsulated in challenge 2: Developing intelligent systems for more adaptive and responsive HCI solutions that cater to diverse user needs and contexts. Publication trends in the Scopus database from 2018 to 2024 reveal a rapid growth in research on agent-based and intelligent HCI systems, as illustrated in Fig. 11. It represents the evolving progress in the Intelligent Virtual Environment (IVE) and agent-based systems from 2018 to 2024 with an upward curve of steadily increasing publications. By presenting the MAMbO5 ontology, it categorizes IVEs into non-virtual (workspace, artifact, agent) and virtual components (human-immersed agents, IVE workspaces). Among the

core contributions are imaging systems, the Quality of Experience (QoE) management, encryption, and scenario fusion for mixed reality (Nguyen et al., 2024; Kasprzak et al., 2020). A fuzzy logic workflow illustrates adaptive decision-making in CPS (Xu et al., 2015). Therefore, it emphasizes improvements in adaptability, security, and HCI concerning IVE (L. Liu, 2022; Karaduman, Tezel, and Challenger, 2021).

Agent-based systems have been applied across diverse domains:

- **Mixed reality and IVE:** Frameworks like the MAMbO5 ontology provide design foundations for agent-based interfaces, while collaborative frameworks address security and distributed processing. Social interaction studies, such as interactions with Siri (S. K. Lee, Kavya, and Lasser, 2021), emphasize trust and social presence in human-AI interactions.
- **Public space and emotional intelligence:** Agent-based models for public space usage (Cheliotis, 2020; Nwokoye, Ejiofor, Orji, and Mbeledeogu, 2016) inform human spatial behavior. Emotional

intelligence is integrated into systems like the emotion-driven reasoning model (J. Wu et al., 2024), improving negotiation success.

- **Healthcare and assistive technologies:** Multi-agent systems have been developed for elderly care, visually impaired assistance (Oliveira et al., 2022), and platforms like EREBOTS (Calvaresi et al., 2021) for personalized behavioral change programs.

Advanced technologies continue to shape the field. In education, multi-agent virtual assistants enhance learning management systems (Laeq and Memon, 2021), while personality modeling in agent-based systems advances interaction customization (Ahrndt, 2019). The integration of large language models into agent-based modeling (J. Chen et al., 2024) introduces new possibilities for learning and decision-making systems. Additionally, intelligent hybrid systems in museums improve UX (Xia and Zhang, 2015) and ergonomic research explores post-dualist sociotechnical systems (Holman et al., 2020). Lastly, a systematic review of decision support systems (Khemakhem, Ellouzi, Ltifi, and Ayed, 2020) consolidates findings from 58 studies, providing valuable insights into the evolution of agent-based HCI systems. This comprehensive categorization highlights the rapid evolution of agent-based systems for HCI, emphasizing their diverse applications and innovative potential across various domains.

3.3.3. Edge computing for real-time adaptivity (Challenge 3)

Edge computing, which plays a crucial role in reducing latency and enhancing real-time processing, becomes even more critical in the context of I5.0 (M. Sharma, Tomar, and Hazra, 2024). By bringing processing power closer to the user, edge computing supports the creation of adaptive HCI systems that can respond immediately to human actions, thereby enabling more natural and efficient human-machine interactions (Sharma et al., 2023). In a smart factory setting, human workers can interact with machinery in real-time, with the system adapting on the fly to their inputs, thus enhancing both safety and productivity. This section will address challenge 3: Integrating advanced computing paradigms to enhance real-time security in HCI systems while maintaining optimal performance.

According to the publication search results from 2018 to 2024 regarding the topic of edge computing AND real-time AND adaptive AND security AND HCI, in the Scopus database, the number of publications shows a rapid growth trend in recent years, as shown in Fig. 12. It highlights how the edge-cloud integration matured over time towards AR/VR applications as per the security and efficiency, which is made by cloud servers, edge servers within an edge-cloud interaction. Data is exchanged through Named Data Networking (NDN) and Wi-Fi networks connected with AR/VR devices (Hossain and Muhammad, 2019; Márquez-Sánchez et al., 2023). The bottom timeline of Fig. 12 depicts the increase in publications, representing the progressively growing field of interest. The three layers (levels 1, 2, and 3) form the hierarchical scalability of distributed ledger technology itself, targeting increased security and privacy for these systems. They cover applications as diverse as smart homes and IoT-enabled cloud applications, thus

showcasing some of the various ways in which edge-cloud synergy has a real-time and immersive context (Y. Huang et al., 2018; Shannigrahi et al., 2020).

As we explore these transformative roles, it becomes essential to examine the specific innovations and applications that are shaping edge computing today.

- **Cybersecurity solutions:** Stsebachova, (2023) introduces a real-time and distributed cybersecurity solution employing dynamic risk assessment and monitoring, which manages adaptive cybersecurity in autonomous vehicles and inspires privacy-preserved processing across other domains. Pešić et al., (2020) propose a comprehensive security framework addressing credibility, authentication, authorization, verification, and integrity, reinforcing secure edge computing practices. Golpayegani et al., (2024) reviews adaptive edge computing systems, identifying research challenges and suggesting strategies to maintain performance in dynamic environments.
- **Collaborative software:** Ortegat et al., (2022) highlights the transition of collaborative software, such as a pictorial game (Margarido, Roque, Machado, and Martins, 2024), from cloud to edge computing, significantly reducing latency and improving scalability. These advancements pave the way for enhanced user interaction and dynamic adaptability in edge computing environments.
- **Architectural innovations:** Márquez-Sánchez et al. (2023) integrate federated learning and deep reinforcement learning into edge computing architectures to optimize energy consumption in smart buildings. Their approach addresses both technical efficiency and user-centric adaptability, demonstrating the potential for edge computing to create more sustainable infrastructures.
- **Smart cities and IoT:** Walczak et al., (2023) assesses the adoption of IoT sensors in smart cities, emphasizing factors such as trust, perceived usefulness, and data security to encourage acceptance. These findings underscore the critical role of edge computing in enabling scalable and efficient IoT networks tailored to diverse urban challenges (Stephanidis et al., 2019).
- **Energy systems:** C. Z. Liu, (2022) analyzes edge computing applications in smart energy systems, addressing key market and technical barriers while providing insights into policy-driven advancements. This work highlights the intersection of energy management and real-time edge computing technologies.
- **Wearable technology:** Covi et al., (2021) proposes low-power adaptive systems utilizing neuromorphic computing for smart wearable devices, ensuring efficient real-time processing. These innovations reflect the broader trend toward personalized and resource-efficient edge computing applications.
- **Healthcare:** Z. Liu and Zhang, (2023) integrates multimedia sensing and blockchain technologies to optimize healthcare supply chain management. This approach combines edge computing with secure data management to enhance efficiency and reliability in healthcare systems.

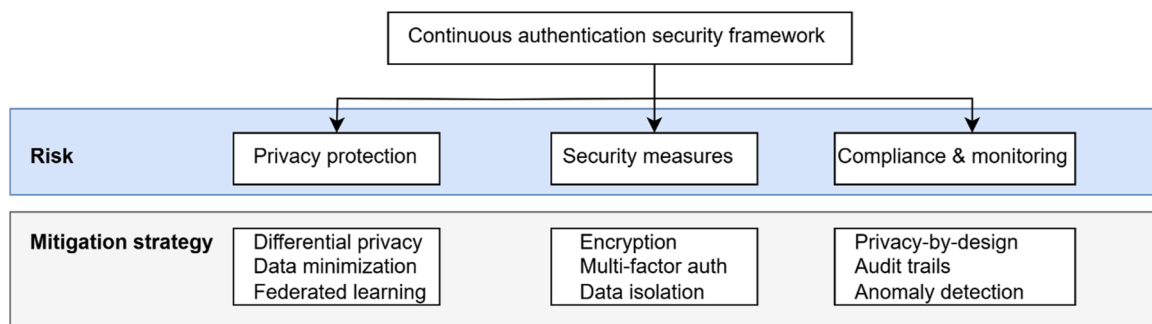


Fig. 10. Primary components of a continuous authentication security framework and their mitigation strategies.

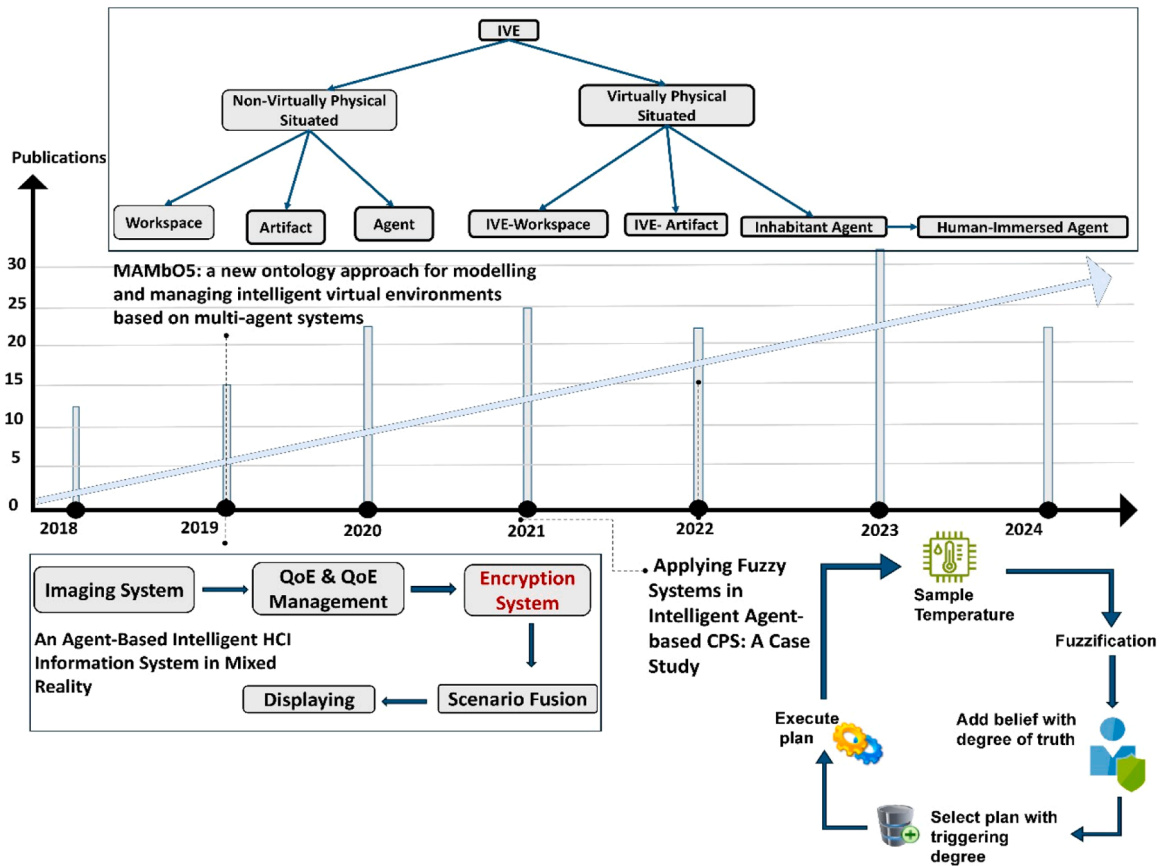


Fig. 11. Evolution of intelligent agent-based HCI systems.

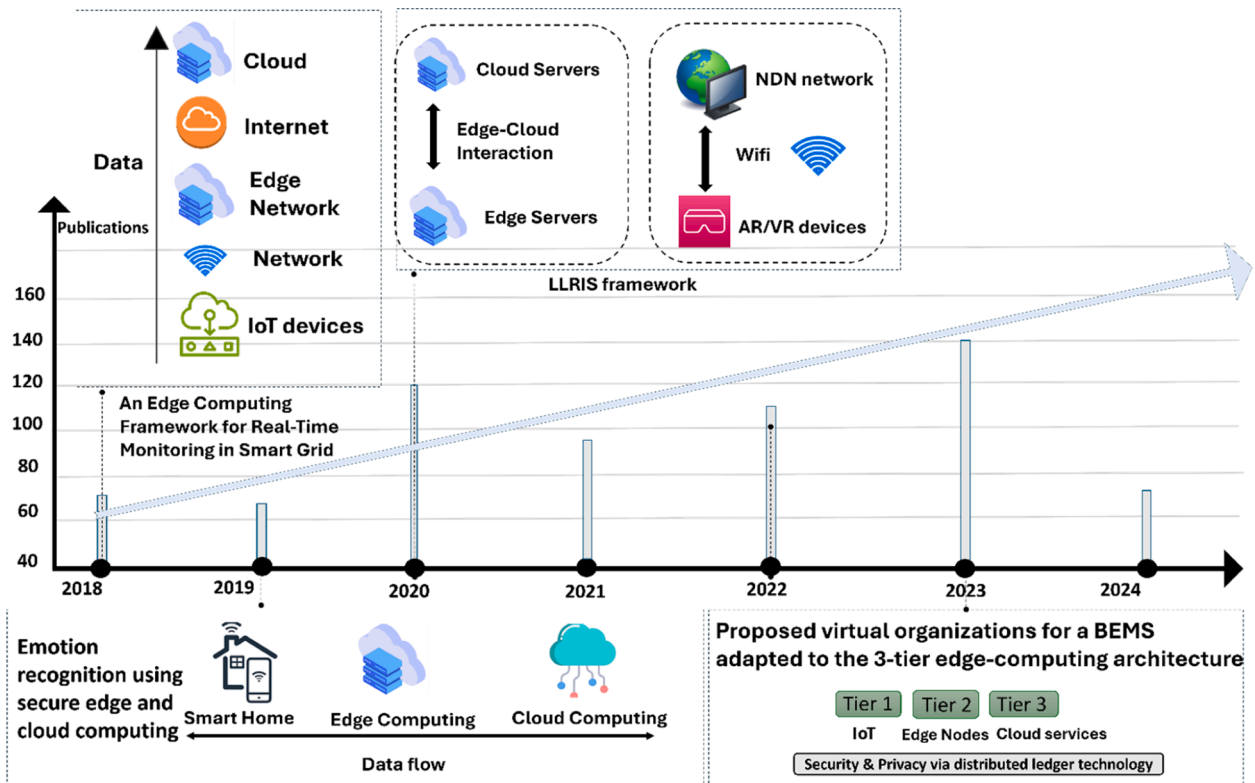


Fig. 12. Evolution of edge computing for real-time adaptivity.

These advancements underscore the versatility of edge computing, paving the way for future innovations while addressing performance, security, and usability concerns. Additionally, Sun et al. (2024) and Fang et al. (2023) enhance facial expression recognition by improving real-time performance with reduced resource usage. Wang et al. (2020) integrates cloud, edge, and fog computing to address big data challenges in CPS (Yin et al., 2023). Radanliev et al. (2020) explore IoT security through cognitive engine design and anomaly detection, while Shannigrahi et al. (2020) introduce the LLRIS framework

for networking. Robai (2024) optimize resource allocation with the MOERA algorithm, and Xiao et al. (2019) review edge computing security, proposing a unified security framework. Robai (2024) also introduce an IoT-cloud architecture with trust evaluation to improve security and efficiency. H. Zhang et al., (2019) presents a hierarchical emotion recognition system with a perturbation-aware defense mechanism. Huang et al., (2018) offers a holistic solution for real-time smart grid monitoring, outperforming traditional methods. Hossain and Muhammad, (2019) propose a system for privacy-preserved processing using secret sharing, ensuring high recognition accuracy in decentralized systems. Finally, Taherizadeh et al., (2018) highlights the need for comprehensive monitoring, emphasizing real-time risk assessment within edge computing frameworks.

3.3.4. Summary of key HCI technologies

Table 3 provides a summary of the key HCI technologies, highlighting their primary advantages and potential disadvantages. These technologies are noted for their advanced capabilities in security, adaptivity, and real time processing, yet each comes with its own set of challenges.

4. Novel methodology and empirical validation of adaptive HCI framework: application to FDD and self-healing within I5.0

Despite the advancements highlighted in the previous sections, current HCI methodologies face limitations when applied to critical systems such as FDD in Smart Manufacturing (SM). These traditional approaches often lack the adaptability and security (PacauxLemoine, Trentesaux, Rey, and Millot, 2017) required to handle dynamic environments, leading to inefficiencies and potential vulnerabilities. Additionally, existing methods do not adequately account for the variability in user behavior, cognitive load, technological proficiency, cultural differences, and accessibility needs which can significantly impact system reliability and UX (Wang and Lajoie, 2023). To address these limitations, a new approach is necessary, one that ensures accessibility, diversity, and user well-being while enhancing system performance, operational resilience, and user satisfaction.

4.1. Integrating human-centric HCI with FDD and self-Healing systems in I5.0

In I5.0, which emphasizes a human-centric approach, the integration of HCI with FDD systems is essential (Kemény, Váncza, Wang, and Wang, 2021). FDD systems monitor operational parameters and detect anomalies early, while self-healing mechanisms automatically take corrective actions, ensuring minimal disruption and downtime (Wang et al., 2024). This fusion of human expertise with advanced FDD systems empowers operators to focus on high-level tasks while relying on the system's automated processes to handle routine monitoring and fault management (Afolalu, Nwankwo, & Azeez, 2024). By addressing the diverse needs of global users, this integration exemplifies the collaborative potential of I5.0, ensuring operational efficiency, adaptability, and resilience.

4.2. Introducing a new conceptual model of human-centric approach in FDD and self-healing systems

This innovative methodology creates a human-centric, adaptive, and secure HCI system for FDD. Following the data flow through the system (Dai and Gao, 2013), integrating advanced technologies like IoT, Machine Learning (ML) (Bi et al., 2024), and real-time cognitive state assessment.

The main contribution of the proposed conceptual model illustrated in Fig. 13 is its transformation of the FDD system from a static, reactive approach into an intelligent, proactive solution (Zeng, Cai, Lin, and Wen, 2022). This data flow, the system continuously learns from data and interactions, ensuring the delivery of relevant information, reducing cognitive load (Ferreira et al., 2014), and facilitating efficient FDD and self-healing in SM environments (Aldrini, Chihi, and Sidhom, 2024).

4.3. Methodology description

This section provides a detailed, step-by-step explanation of the proposed conceptual model, highlighting the flow of data and the interconnected processes. Each step is carefully described to illustrate how the system integrates adaptive features, user parameters, and real-time environmental data to achieve a seamless and human-centric solution. By following the data flow, this section demonstrates how the model transitions from supervised processes to fault detection, diagnosis, and self-healing, culminating in an adaptive interface that supports diverse user needs and operational contexts.

4.4. Block 1: Supervised process

Beginning with the supervised process, data is collected from various sources in the manufacturing environment.

Data Collection and integration (Step 1): Actuators, processes, and sensors form the core of this block. Smart sensors (like S1) continuously monitor machine (M1) parameters such as temperature, vibration and energy consumption. This data, along with maintenance records and technician reports, is collected in real-time (Fatoba, Jaiyeoba, Oladosu, & Oyewole, 2024). To ensure efficiency and precision, the system dynamically adjusts sampling rates based on operational conditions, prioritizing critical data during potential fault scenarios while reducing the load during normal operations. For example, in a noisy environment, the system prioritizes vibration data over acoustic signals to maintain data quality. Simultaneously, environmental factors such as noise levels, lighting, and temperature are monitored to optimize data collection and workflows.

An important data layer is needed which is the user data. Operators must input their technological skills and accessibility preferences, allowing the model to later adapt workflows and actions to individual capabilities. During onboarding, operators specify their technological proficiency, accessibility preferences, and interaction modes. This user-centric data is combined with sensor and environmental data to enable highly personalized workflows, while behavioral patterns, such as frequent errors or task preferences, are logged to refine system recommendations over time. Collected data undergoes rigorous preprocessing, including cleaning, normalization, and annotation, to ensure consistency and readiness for downstream analysis. Cultural and regional metadata, such as language preferences and region-specific maintenance norms, are appended to align workflows with diverse user needs. The system's robust security framework protects data integrity through advanced encryption standards for data at rest (Alemami et al., 2023), transport layer security for data in transit (Meier, 2023) and implement role-based access control (Butt et al., 2023) and multi-factor authentication (Suleski et al., 2023).

Knowledge graph construction (Step 2): As preprocessed data flows from step 1, it is used to construct and continuously update a comprehensive knowledge graph (Schmidt, RinconYanez, Kharlamov, and

Table 3
Advantages and disadvantages of key HCI technologies.

Technology	Advantages (+)	Disadvantages (-)	Challenge 1	Challenge 2	Challenge 3
Behavioral biometrics	√Enhanced security√Nonintrusive√Adaptable to user behavior	×Variability in behavior×Privacy concerns	√		
Intelligent agent-based HCI	√Adaptive√Personalized UX√Cognitive state assessment	×Challenges in real-world validation×Complexity in modeling accurate cognitive states		√	
Edge-computing for real time adaptivity	√Low latency√Real-time data processing	×Security vulnerabilities×Scalability issues			√

Paschke, 2024). This graph links operational parameters (e.g., sensors, machines, and faults) with user data (e.g., proficiency levels, behavior patterns, and accessibility needs) and contextual metadata (e.g., environmental and cultural factors). NLP techniques extract entities (like sensors, machines, faults) and their relationships from the incoming data (Hirschberg and Manning, 2015). For example, the graph may link a sensor measuring vibration on Machine M1 to historical overheating issues and maintenance actions, while tagging operators experienced in similar faults for potential assignments. The knowledge graph also supports multilingual labels and region-specific instructions, making it accessible to operators worldwide. This enriched graph becomes the

backbone for FDD, allowing the system to make decisions tailored to the unique contexts of both machines and users. This graph is secured using integrity checks and maintains detailed audit logs (Li et al., 2023).

4.5. Block 2: Fault detection and diagnosis

As data flows from the supervised process, it enters the FDD block, where anomalies are identified, isolated, and addressed. This block dynamically adapts based on the knowledge graph and user parameters.

Fault detection: The system continuously analyzes the incoming data streams, comparing operational patterns with the knowledge stored in

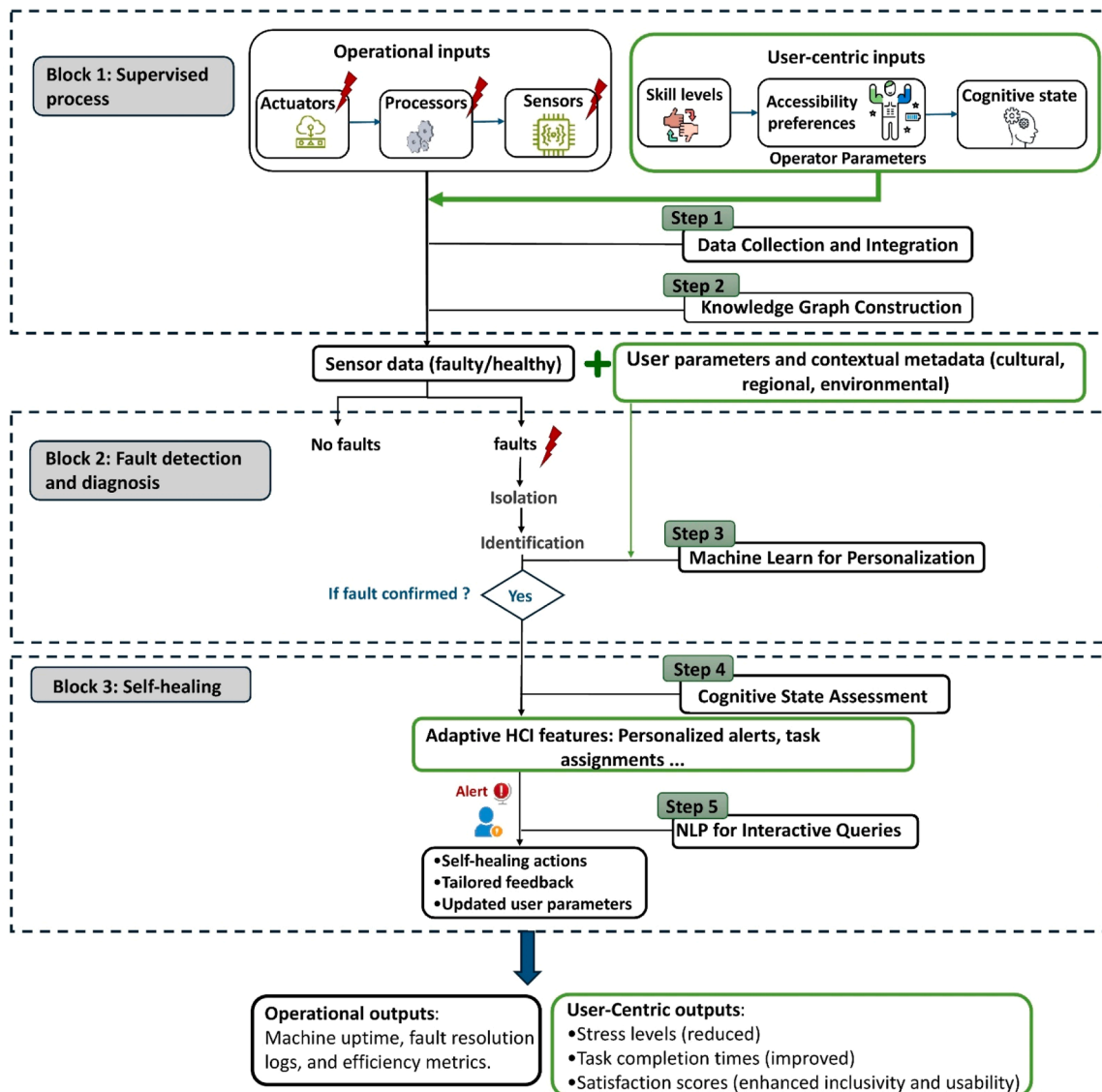


Fig. 13. New conceptual model of human-centric approach in FDD and self-healing system.

the graph. It employs adaptive ML models that learn from both the data and operator interactions, enabling real-time fault detection with high accuracy and reduced false positives (George et al., 2024; Yuan, He, Qin, Zhao, and Zhang, 2021). For instance, if sensor data indicates unusual vibrations in M1, the system correlates it with similar past incidents in the knowledge graph to confirm whether it constitutes a fault. Alerts are customized based on operator profiles: a novice technician might receive step-by-step visual guidance and tutorials, while an experienced operator is presented with a concise fault summary and advanced resolution options. Stress and cognitive load assessments further personalize the response; during high-stress periods, the system simplifies alert information to focus on immediate actions, minimizing decision fatigue. If no faults are detected, the data continues to flow, updating the knowledge graph and refining the ML models. However, when a fault is detected, the system moves to isolation and identification phases.

Fault isolation and identification: Once a fault is detected, the system isolates the affected components or processes and determines the fault's nature. Using the knowledge graph, it identifies potential causes and suggests context-specific resolutions. For example, in tropical regions, overheating issues might prompt airflow optimization as a priority step. Complex faults trigger collaborative dashboards, where tasks are dynamically distributed among team members based on their expertise, accessibility needs, and cognitive states. For example, an operator skilled in electrical troubleshooting may be assigned diagnostic tasks, while a trainee is tasked with simpler support activities.

ML for personalization (Step 3): If a fault is confirmed, the system activates its personalization features. It develops and updates user profiles based on interaction history and analyzes behavior patterns (Ogidan et al., 2025). For instance, if Technician T1 has efficiently handled overheating issues in the past, the system might prioritize alerting T1 for this type of fault. This step employs advanced security measures like k-anonymity, differential privacy, and federated learning to protect user data while still benefiting from global model improvements (Duan et al., 2023).

4.6. Block 3: Self-healing

Once a fault is identified and personalized, the data flows into the self-healing block.

Cognitive state assessment (Step 4): As the system prepares to alert operators, it first assesses their cognitive state using physiological sensors and interaction metrics (Ba et al., 2023). High stress or fatigue prompts adjustments in workflow complexity. For example, during a peak stress period, the system may reassign tasks or provide simplified, calming visuals to help operators focus. ML models predict the operator's cognitive load, which influences how information is presented (Wang et al., 2023). For example, if a technician's stress levels are high, the system might simplify the alert display or suggest task reassignment. Accessibility features ensure inclusivity during self-healing processes. Operators with visual impairments can rely on braille alerts and screen readers, while voice commands provide hands-free interaction. Multi-sensory feedback, such as vibrations for critical alerts, further enhances usability across diverse operator groups. Physical accessibility needs are considered as well: voice commands enable hand-free interaction, screen reader compatibility ensures usability for visually impaired users, etc... This sensitive data is protected using encryption and privacy-preserving analysis techniques (Munjal et al., 2023).

The system generates tailored alerts, delivering them in formats aligned with operator preferences and accessibility needs. Alongside alerts, automated self-healing actions are initiated based on historical resolutions stored in the knowledge graph. For example, if an overheating fault is detected, the system might automatically increase cooling fan speeds and notify operators of the action taken. As the system is localized, culturally relevant instructions accompany self-healing workflows. For instance, repair suggestions may refer to spare parts commonly available in the operator's region. Gamified tutorials guide

operators through manual interventions, fostering confidence and skill development.

NLP for interactive queries (Step 5): As operators respond to alerts, they can interact with the system using natural language queries. The system extracts relevant information from the knowledge graph to provide contextual responses (Yang et al., 2023). For instance, an operator might ask, "When was the last maintenance check on M1?" and receive an immediate, relevant answer. Localized interfaces support multiple languages, enabling accessibility for diverse user groups, along with dynamic switching that allows operators to change languages without losing operational context. This communication is secured using transport security, digital signatures, and robust authentication (Bagherian et al., 2024).

4.6.1. Key features of the adaptive HCI and their applications

To comprehensively address the diverse needs of operators and enhance the efficiency and inclusiveness of the system, the proposed conceptual model incorporates a range of adaptive features. These features are derived from the methodological processes described earlier, integrating insights from supervised processes, fault detection, and self-healing workflows. Table 4 summarizes the key features, categorizing them based on their purpose and operational focus, while highlighting their practical applications and examples.

This categorization shows that the adaptive features offered make HCI interfaces more accessible, personalized, and functional in diverse contexts. However, their implementation on a large scale, however, presents few challenges, especially in such regions as technology integration, cost, and flexibility in meeting varied user demands. Consequently, the realization of such an expectation would necessitate the testing of these features in field conditions, the refinement of their designs based on user feedback, and enhancement of their scalability.

4.7. Empirical validation of the adaptive HCI proposed model

The purpose of this empirical study is to validate the adaptive HCI model proposed for improving operator well-being, task efficiency, and inclusivity within a simulated manufacturing environment. By integrating cognitive accessibility, dynamic task assignment, personalized workflows, and collaborative dashboards, the study aligns with I5.0 principles and showcases the adaptability of the model under diverse operator conditions and fault scenarios.

The study emphasizes: (1) Stress reduction through adaptive task flows, (2) cognitive load management using tailored tutorials, (3) task fairness via dynamic workload balancing, (4) accessibility satisfaction through ergonomic tools and (4) reduced fault recovery times using self-healing actions. Fig. 14 illustrates the workflow adopted to validate the proposed approach, detailing the steps from environment setup, implementation of the adaptivity algorithm, and subsequent simulation of the operator's response. After the implementation, the results for this empirical study were derived using a systematic approach involving simulated data generation, operator behavioral modeling, and quantitative performance analysis. In the following sections, each step of the workflow is detailed.

4.7.1. Environment setup

To ensure realistic industrial conditions, machine data, operator profiles, and fault scenarios were generated using simulation tools such as MATLAB and Python frameworks (e.g., SimPy and OpenAI Gym). These tools allowed the creation of dynamic, real-time data streams based on predefined probabilistic models and behavioral assumptions:

- Machine Parameters:
 - Temperature (70°C–120°C): Overheating was simulated when temperatures exceeded 110°C.

- Vibration: Abnormal spikes were modeled to represent faults caused by worn-out components.
 - Pressure (40 psi to 15 psi): Simulated pressure drops trigger fault scenarios.
 - Noise levels (60–90 dB): Increased noise was used to test accessibility tools like text-to-speech alerts.
- Operator Profiles:
 - Simulated 40 operators with diverse skill levels, cognitive loads, and accessibility needs.
 - Stress levels were modeled using Heart Rate Variability (HRV), ranging between 60 bpm (low stress) and 120 bpm (high stress).
 - Behavioral responses were simulated based on performance probabilities (e.g., error rates increase under cognitive overload).
 - Fault Scenarios:
 - Overheating (M1): Modeled rising temperatures to trigger workflow adaptations.
 - Abnormal vibration (M2): Simulated faults through vibration spikes exceeding safety thresholds.
 - Pressure drop (Hydraulic system): Simulated sudden pressure loss disrupting production.

4.7.2. Adaptive algorithm implementation

The adaptive algorithms underlying the HCI model were systematically implemented by aligning with the methodology description,

integrating **supervised processes**, **FDD**, and **self-healing functionalities**. Each process was designed to ensure efficient, personalized, and human-centric operations.

4.8. Block 1: Supervised processes

The foundation of the adaptive system was established through robust supervised processes, emphasizing real-time data collection, integration, and contextual understanding:

Data collection and integration: Sensor data was processed in real-time using python libraries like pandas and scikit-learn. Preprocessing steps included normalization, cleaning, and feature extraction to ensure high-quality inputs for downstream tasks. To reduce data load, dynamic sampling rates were employed, prioritizing vibration analysis during high-noise conditions while scaling down sampling during routine operations.

Knowledge graph construction: Neo4j and python's networkx library were used to build a dynamic knowledge graph that interlinked:

- **Machines (M1)** with historical fault data.
- **Operator profiles**, capturing skill levels, stress patterns, and behavioral responses.
- **Contextual metadata**, such as environmental and cultural parameters. NLP tools like spaCy enriched the graph by extracting fault patterns from sensor data and associating them with historical resolutions.

Table 4

Categorization and real-world applications of the proposed adaptive HCI features.

Feature category	Feature	Description	Example	Real-world application
Accessibility needs	Cognitive accessibility	Simplified workflows and intuitive layouts adapt based on stress and cognitive load.	A crisis response system limits non-essential information during high-stress scenarios.	Emergency management systems that prioritize critical tasks during natural disasters.
	Physical accessibility	Compatibility with assistive devices and hands-free operation.	Voice commands for initiating actions and haptic alerts in noisy factory environments.	Assembly line management in factories with noisy or hazardous conditions.
	Ergonomic customization	Configurable input devices for mobility needs.	Mobility-impaired operators use touchless controls in a healthcare setting.	Control panels in hospitals for surgeons and mobility-impaired staff.
Technological proficiency	Workflow customization	Tailored guidance for novices; advanced analytics for experts.	Novices get step-by-step guidance; experts access raw data in SM dashboard.	Industrial automation systems provide real-time data for engineers and guidance for trainees.
	Training personalization	Adaptive training modules with gamified learning paths.	Scenario-based tutorials simulate fault resolution in an IT operations center.	Cybersecurity centers train analysts for intrusion detection and response.
Collaboration	Dynamic task assignment	Tasks distributed based on skill levels and cognitive states.	Skilled operators handle complex diagnostics; support tasks assigned to less experienced users.	Customer service centers assign calls based on agent expertise and stress levels.
	Collaborative dashboards	Role-specific dashboards for task transparency and accountability.	Operators track progress and share insights in a space exploration mission control room.	Space mission operations, where engineers and scientists monitor spacecraft systems.
	Cross-functional collaboration	Interfaces translate technical data for non-technical users.	Maintenance and management teams collaborate with localized interfaces in a global logistics hub.	Global supply chain management platforms integrating technical and business insights.
Behavioral and cultural adaptivity	Behavioral insights	Tracks interaction patterns to refine workflows in real time.	Prioritized guidance for operators struggling with recurring faults in a retail inventory system.	E-commerce warehouse management refining inventory tracking workflows.
	Cultural adaptivity	Localized interfaces with region-specific workflows.	Maintenance steps referencing locally available spare parts in a mining operation.	Mining operations using location-based equipment references for efficient repair.
	Multilingual design	Supports multiple languages with dynamic switching.	Operators switch languages in a multinational call center without workflow interruptions.	Multinational customer support teams handling inquiries in multiple languages.
Personalization	Tailored workflows	Adapts workflows and alerts based on operator profiles.	Novices see simplified alerts; experienced operators access advanced summaries in a warehouse system.	Automated warehouse systems provide role-based alerts for workers and supervisors.
Environmental adaptivity	Dynamic interface adjustments	Interfaces adjust to lighting and noise conditions.	Text-to-speech activated in noisy construction sites for improved communication.	Construction management tools ensuring usability in varying environmental conditions.
	Multimodal interaction	Supports diverse interaction modes like voice commands and touch interfaces.	Hands-free navigation for emergency responders using voice commands.	Firefighting equipment interfaces allowing voice-controlled navigation during rescues.

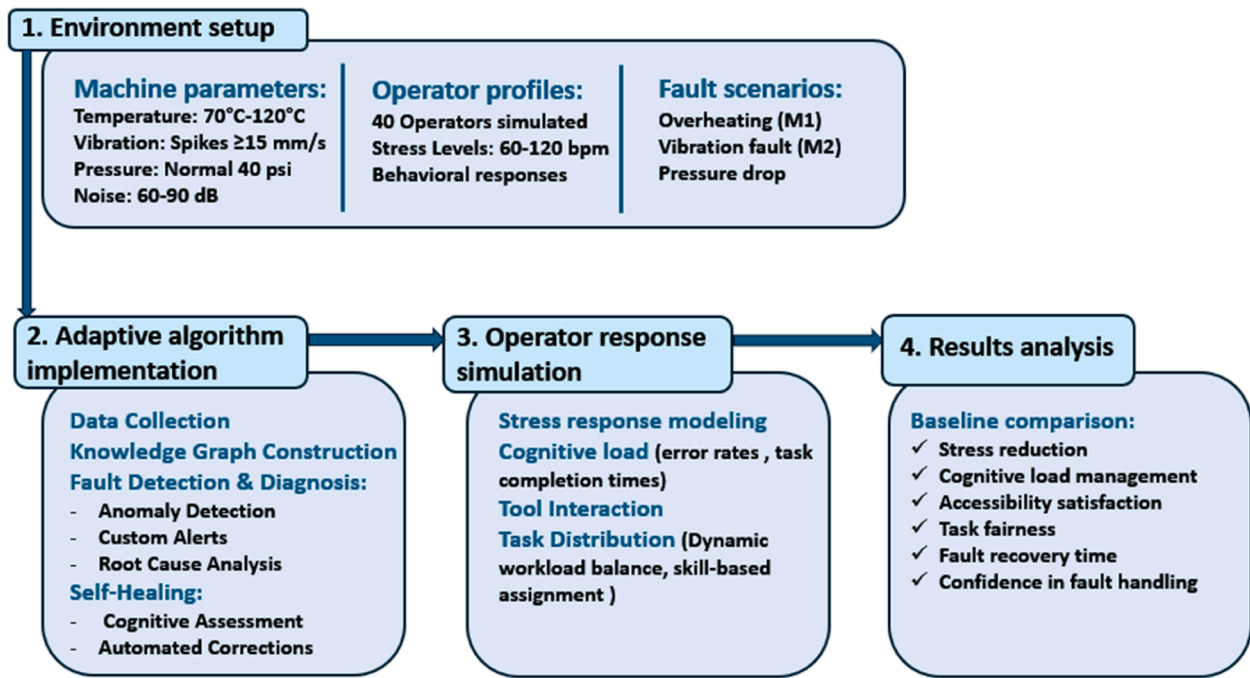


Fig. 14. Workflow of the empirical validation development.

4.9. Block 2: Fault Detection and Diagnosis

Adaptive ML models facilitated real-time fault detection, isolation, and operator-specific responses:

- ✓ **Fault detection:** ML models, including isolation forests from scikit-learn, identified operational anomalies by detecting deviations from predefined thresholds. Alerts were tailored to user expertise:
- ✓ Novices received step-by-step instructions for troubleshooting.
- ✓ Experts were presented with concise fault summaries using pre-trained transformer models (Hugging face).

Fault isolation and identification: Root causes of faults were pinpointed by cross-referencing knowledge graph insights with live sensor data. Tasks were dynamically distributed via collaborative algorithms:

- ✓ Operator availability, skill levels, and stress indicators influenced task assignments.
- ✓ Redistribution using scheduling libraries (joblib) to maintain equitable workloads.

Personalization: User profiles were updated dynamically based on interaction patterns using reinforcement learning (stable-baselines3). Using TensorFlow, the algorithm ensured user privacy while optimizing global ML algorithms, supporting scalable personalization across diverse environments.

4.10. Block3: Self-healing functionalities

The self-healing component complemented fault detection by automating corrective actions and mitigating operator fatigue:

Cognitive state assessment: Physiological indicators such as HRV were analyzed using python's statsmodels to evaluate stress and fatigue levels. Adaptive workflows were activated:

- ✓ Simplified task interfaces during high-stress scenarios.
- ✓ Reassignment of tasks to operators in less fatigued states.

Automated self-healing actions: Corrective measures, such as

increasing fan speeds to mitigate overheating, were executed using MATLAB control algorithms. For manual interventions, gamified tutorials developed with python's pygame library enhanced user engagement and confidence.

Integrated Workflow: The workflow unified data-driven fault detection with operator-centric adaptive responses:

Fault detection: Real-time sensor data (e.g., temperature, vibration) was monitored against predefined thresholds using rule-based anomaly detection algorithms. Faults were linked to relevant historical solutions and operator expertise via the knowledge graph.

Operator-centric responses: Adaptive responses, including personalized alerts and task redistribution, were triggered based on the operator's profile and stress state. Tools like scipy for data analysis and pandas for data integration supported seamless implementation.

4.10.1. Simulating operator responses

The adaptive HCI system was rigorously evaluated using behavioral and cognitive models that simulated operator interactions under varying conditions. These simulations were designed to measure performance across stress levels, cognitive load, accessibility satisfaction, and task fairness, providing insights into the system's effectiveness in dynamic, fault-prone scenarios. The results offer a comprehensive understanding of how adaptive workflows can improve operator performance and system efficiency. Once faults are detected, adaptive responses are generated based on the operator profile and stress state. These adaptive measures were critical in maintaining system efficiency during faults and high workload scenarios. These simulations provided actionable insights into how operators interacted with the system during high-stress, high-load scenarios and how the system responded adaptively to mitigate challenges. This section highlights the simulated behaviors and technologies used to assess the adaptive HCI system's effectiveness under various operational scenarios systems.

4.11. Stress reduction

Stress reduction was evaluated using HRV data, which was monitored before and after tasks to measure stress responses. HRV metrics were calculated from raw physiological data using pyhrv. Gaussian

distributions were used to model the variability in stress levels for tasks performed with and without adaptive workflows. The results showed that HRV remained high (~ 110 bpm) in traditional workflows but was reduced by 30 % in adaptive workflows, indicating a significant decrease in stress levels. Simplified workflows also enabled faster task completion, further correlating with reduced stress.

4.12. Cognitive load management

Cognitive load management focused on error rates, task completion times, and perceived workload. Error rates were simulated for novice operators under guided and non-guided workflows. Monte Carlo simulations revealed that adaptive tutorials reduced error probabilities by 25 %, demonstrating the benefits of guided workflows. Additionally, time logs showed that novices performing tasks with tutorials completed them significantly faster than those without guidance. Surveys captured perceived workload, with operators reporting lower cognitive load after completing tasks using adaptive workflows. Statistical analysis of survey data with `scipy.stats` confirmed these observations.

4.13. Accessibility satisfaction

Accessibility satisfaction was another critical area evaluated. Interaction data tracked the use of touchless controls and text-to-speech alerts during simulations, particularly for mobility-impaired operators. High-noise environments (above 90 db) triggered text-to-speech alerts, which increased clarity and usability. These features were powered by speech recognition for voice commands and `pyttsx3` for text-to-speech synthesis. The results showed that 85 % of interactions used touchless controls, and satisfaction with accessibility features improved by 25 %, demonstrating the system's adaptability to diverse user needs.

4.14. Task fairness

Task fairness was measured through workload distribution logs and survey results. Logs from collaborative dashboards tracked task redistribution based on operator skill, stress levels, and availability. Tasks were dynamically redistributed using algorithms implemented with `networkx` to ensure equitable workload. Fairness was measured by deviations from uniform distribution, and the results showed a 20 % improvement in perceived fairness. Surveys validated these findings, with 85 % of operators reporting equitable task distribution after simulations.

4.15. Fault recovery time

Fault recovery time was another area of significant improvement. Time logs were analyzed to extract fault recovery durations, and adaptive workflows triggered self-healing actions to reduce delays. Fault detection and resolution timestamps were processed using `pandas`, revealing that traditional systems resolved faults in 35 minutes, while adaptive workflows reduced this to 20 minutes, achieving a 40 % improvement in fault recovery time.

4.16. Confidence in fault handling

Confidence in fault handling was evaluated through pre-task and post-task surveys, as well as task performance metrics. The simulation of confidence rate levels are higher after completing tasks with adaptive tutorials, with a reported 28 % increase in confidence. Additionally, task logs showed higher success rates and faster completions under adaptive workflows. Survey data was analyzed using `scipy.stats`, and task logs were processed with `pandas` to benchmark these improvements.

4.17. Ensuring reliability of findings

A structured results extraction process ensured the reliability of these findings. Data aggregation integrated machine and operator response data into structured datasets, which were then subjected to quantitative analysis using statistical methods to compare traditional and adaptive workflows. Finally, the results were validated by cross-referencing them against baseline expectations for traditional systems, confirming the advantages of the adaptive HCI model across all evaluated metrics.

These results demonstrate the broad benefits of adaptive workflows in improving operator performance, reducing stress and cognitive load, enhancing accessibility, ensuring fairness, and building confidence in fault handling.

4.17.1. Results analysis

The results were systematically generated and extracted by combining simulated machine data, operator behavioral models, and probabilistic performance analysis. Using simulation tools like MATLAB and python frameworks, realistic fault scenarios and modeling operator responses to evaluate the adaptive HCI system's effectiveness. The findings illustrated in Table 5 demonstrate significant improvements in stress reduction, cognitive load management, task fairness, accessibility satisfaction, and fault recovery time. This structured, data-driven approach ensures repeatability, robustness, and alignment with I5.0 principles, establishing the adaptive HCI model as a scalable solution for human-centric industrial environments.

The proposed methodology represents a significant step toward creating a truly user-centric approach to fault detection, diagnosis, and self-healing in industrial environments. By integrating user parameters into the system, the model ensures personalized, context-aware interactions that enhance user engagement and decision-making. However, while the conceptual framework is robust, achieving such seamless integration in real-world applications presents several challenges (Kunjir, 2024). From addressing the technical complexities of IoT and ML implementation to ensuring scalability, security, and workforce adaptability, overcoming these barriers is essential for the model's success. The following section explores these challenges in depth and proposes actionable strategies to ensure the model's effective deployment and alignment with I5.0 principles.

5. Challenges, mitigation and future trends

5.1. Real-world application challenges of the proposed approach and mitigation strategies

To provide a technical analysis of the challenges and propose convincing mitigation strategies, this discussion emphasizes specific technical challenges and actionable solutions tailored to the implementation of the proposed model in industrial environments. Fig. 15 provides an overview of the technical challenges, in blue, and the ethical challenges, in orange, introduced by the proposed model and the corresponding mitigation strategies, which are analyzed here in detail.

5.1.1. Technical challenges

• High costs and infrastructure requirements

The supervised program conducted in Block 1 is a dynamic data collection, online monitoring, and secure storage application for the handling of big data traffic from sensors, actuators, and user inputs collection. While these systems are needed in actualizing the system objectives, it poses challenging financial constrictions in legacies-constrained environments. Smart sensors, edge computing devices, and secure data repositories would require huge investments, and their interfacing with the existing setups would require an expensive interface between the older equipment and the new ones-implicating hardware replacement.

Modular systems can solve the above problems by employing their architecture allowing incremental updates without the need to replace the whole system. Prepare edge computing devices with pre-installed connectivity bridges such as Message Queuing Telemetry Transport (MQTT) link these new developments to legacy interface systems, thereby saving upfront capex. TensorFlow Lite as well as Apache Kafka provide a free open-source platform for cheap scalability, whereas Low-Power Wide-Area Networks (LPWANs) such as Long Range Wide Area Network (LoRaWAN), can cut communication costs by reducing power needs for distance transmission. Therefore, any case would allow the system to stay with great performance within budgetary constraints.

• Technical expertise and workforce adaptation

Flexible and adaptive features, as elaborated in Block 1 and 2 above, such as customization of workflows and the security of sophisticated ML models will require a setup of a work force capable of handling very complex combinations of IoT and ML. In fact, for inputting accessibility preferences and technology skills, it has been engineered to ensure personalized delivery of workflows by the system. But such a model really becomes unsustainable in low-tech environments as it may lead to errors or inefficiencies.

Deployment and training of ML models can be made easier through pre-built automation tools such as Amazon Web Services (AWS) SageMaker or Google AutoML. This way, operators can interact with the system without programming knowledge using low-code or no-code platforms like Node-RED or Microsoft Power Apps. Moreover, continuous learning programs using platforms such as Coursera or Udemy will enhance the workforce to meet the technical needs of the model. Certification through gamification could motivate the operators further and promote self-confidence in system adaptive features' usage.

In Step 2, the model creates an online knowledge graph (Ye et al., 2024) that combines data from sensors, maintenance logs, and user inputs. However, because of the disparate formats, styles and protocols in which the huge volume of legacy information is available, it can severely limit the data collection in conformity with the knowledge graph. For example, old legacy protocols such as Modbus or others will be unable to meet the need for the creation of a standardized, real-time data-integration mechanism due to their specific proprietary formats. Therefore, a solution to the prior discussion is the deployment of semantic middleware tools such as: Apache NiFi or Talend.

• Data volume, management, and real-time processing

The supervised methodology in Block 1 changes the sensor sampling rates dynamically when fault scenarios are detected, resulting in a huge amount of data that can be processed and analyzed in real time. Meanwhile, Block 2 uses such data for accurate FDD. The process faces a huge challenge to manage the data along low latency.

Edge AI can, therefore, offer a solution to this problem by bringing large-scale preprocessing of data and local analyses in edge devices using frameworks like TensorFlow Lite or NVIDIA Jetson, therefore

saving bandwidth and leading to faster response times; using data compression algorithms such as Snappy or LZ0 along with filtering technology such as Kalman filters further helps in making data management slim without losing value: high-priority data can thus easily and immediately channel through a hybrid cloud-edge architecture (Zhang and Jiang, 2024), all other data going to be analyzed over a long time period, out to the cloud. Tools such as Kubernetes orchestrate this architecture to bring scalability and efficiency.

• Cybersecurity and privacy concerns

The question of cybersecurity and privacy is highly impacted by the model's user-centered design and cognitive state assessment, accessibility-needs and user behavior patterns existing in steps 3 and 4. Sensitive information, such as indication of stress levels or accessibility needs of the operators, might be misused if not properly safeguarded. Additionally, real-time transmission of sensor data and tailored workflows further increases this surface of attacks and make the system more susceptible to breaches.

Zero-trust architecture enforces strict identity verification and access control at every layer to avoid such risks, ensuring only the authorized users access sensitive data. For example, dynamic, role-based access can be managed using Azure Active Directory. Blockchain technologies like Hyperledger Fabric ensure secure and immutable audit logging of data exchanges to ensure data integrity and traceability. Advanced Encryption Standard 256-bit (AES-256) techniques can protect data at rest and during transit. Federated learning frameworks, such as TensorFlow Federated, would allow ML models to learn from decentralized datasets resulting in enhanced performance of the system while maintaining user privacy.

• Scalability and resource constraints

The emulation model's adaptive interface must properly adjust to the dynamics of the workflows depending on the profiles of the users and system states that it will have scaled to support a broader deployment across multiple facilities. The computational demands of real-time fault detection, knowledge graph updates, and personalization features place an immense burden on systems resource, affecting their reactivity.

To this end, serverless architectures like AWS Lambda or Google Cloud Functions allocate computational resources dynamically that are scaled to suit the needs of cost-efficient scaling. MA model compression techniques like pruning, quantization, and knowledge-distillation aid in optimizing models for low resource consumption while not impacting accuracy. Their efficient resource management and workload distribution ensure system responsiveness and scalability, even in resource-constrained environments, through containerization and orchestration platforms such as Docker-kubernetes.

5.1.2. Ethical challenges

• Privacy, consent, and fairness

One of the most serious ethical implications of integrating user parameters like accessibility preferences and cognitive states is with respect to privacy, informed consent, and fairness. Continuous monitoring of operators' physiological data like stress levels raises issues around workplace surveillance, privacy, and misuse of personal data. It is a tough balancing act between using data to make personalization great and protecting people's privacy. Moreover, if the training dataset is not representative of diverse user groups, the chance of biased judgment is high. This may lead to unfairness towards some and exclusion of individuals or even the maintenance of an already inequitable workplace.

These may be avoided by first and foremost focusing on making things such as transparency and control by the user. Extensive mechanisms for consent would allow the user to fully understand the purpose of the data shared with others and to be able to withdraw that consent at any time. Annual audits of the system would be essential to examine any possible disparities and ensure that those

Table 5
Comparative results analysis of adaptive vs. traditional HCI systems.

Metric category	Traditional system (baseline)	Adaptive system	Improvement
Stress reduction	HRV: 110 bpm	HRV: 77 bpm	30 % reduction
Cognitive load	Error rate: High	Error rate: Reduced	25 % reduction
Accessibility	Limited accessibility	85 % adoption	High adoption
Task fairness	Uneven workload	Equitable workload	20 % improvement
Fault recovery time	35 minutes	20 minutes	40 % faster resolution
Confidence	Low confidence	High confidence	28 % increase

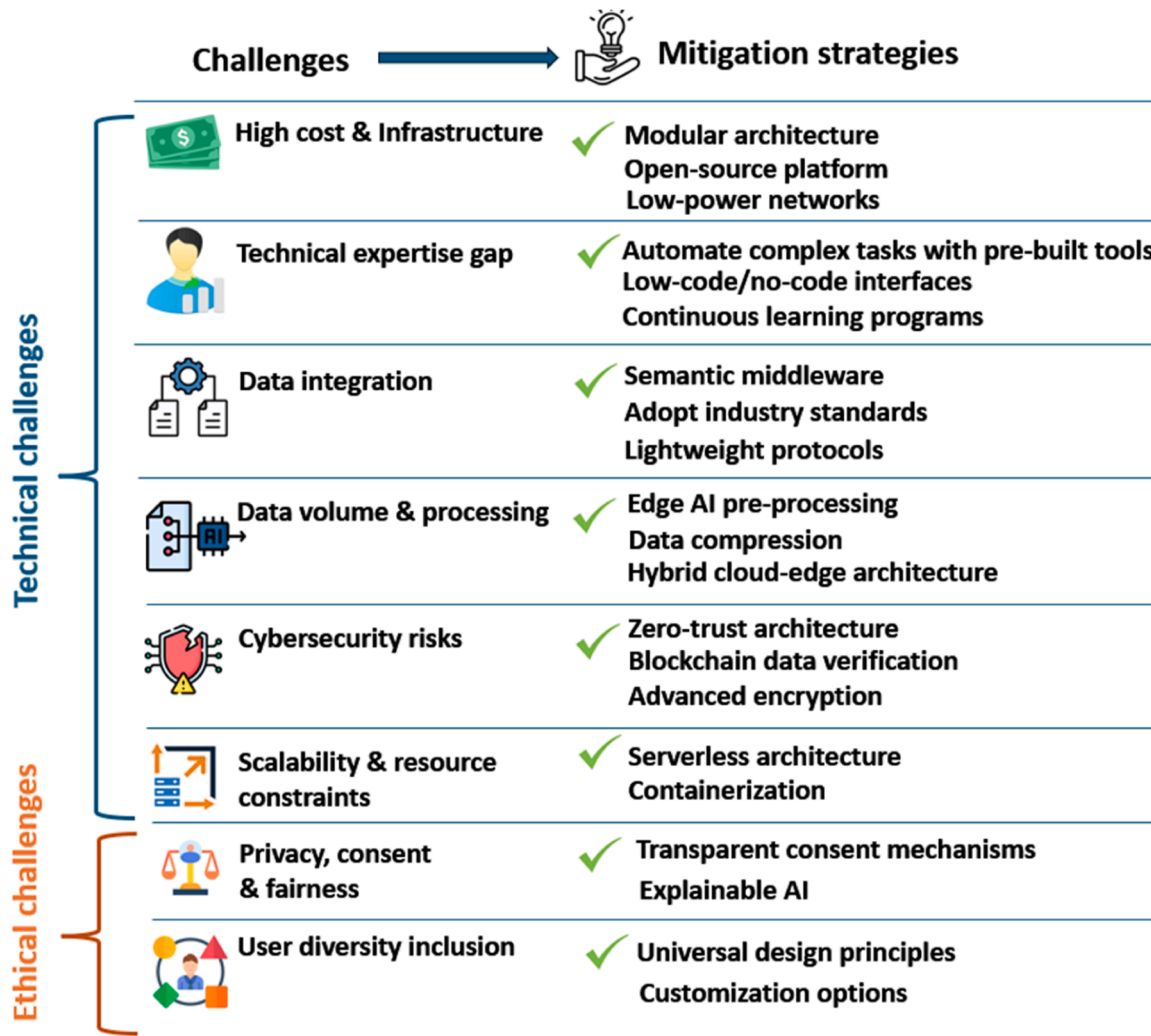


Fig. 15. Real-world application challenges and mitigation strategies.

disparities are addressed so that the model will treat all users fairly, regardless of personal background or demographic characteristics. Moreover, the system needs to impose accountability through a clear explanation for its decisions, thus making operators trust its recommendations while keeping their own agency. Embedding all these ethical mechanisms makes the model fair and privacy-protected as well as trust-building for the users.

- **User diversity inclusion**

Creating adaptive and personalized interfaces is meant for a wider variety of users with different needs, cultural backgrounds, and levels of technology literacy. Designing a genuinely inclusive system, however, encounters great ethical challenges because interfaces must balance localized and culturally acceptable designs with the danger of oversimplifying or stereotyping regional norms; accessibility devices such as screen readers and commands might be implemented to ensure that such devices are genuinely helpful and create lives without visually impaired persons instead of being merely taken on as an addition which fails in practice.

Ethical inclusivity goes beyond technical solutions, it requires a commitment to valuing and respecting all users. This means actively involving diverse groups, including individuals with disabilities and underrepresented communities, in the design and testing of the system. Such engagement ensures that their voices are heard and that the system

reflects their needs and experiences. Additionally, customization options should empower users to tailor the interface to their preferences, giving them control over their interactions. Recognizing and accommodating the full spectrum of human diversity, whether cultural, physical, or cognitive, is not just a design challenge but a moral imperative. By fostering inclusivity and accessibility, the model can uphold the ethical principles of dignity, equity and respect for all individuals.

5.2. General challenges

Integrating AHCI in SM is an important catalyst for enhanced production, reduced errors, and improved satisfaction with the operators. However, realizing such advances has faced serious challenges in technology, human issues, organization, ethics and economics. Technically, barriers to seamless operation include issues such as data from heterogeneous sources, near real-time response, and system interoperability. Human-centered issues include user adaptability, trust in automated support systems, and usability for various operator profiles. Organizationally, operators face workforce training-related difficulties, adoption resistance, and integrating new systems into existing workflows. Ethical concerns regarding data and cybersecurity, as well as accountability for decisions made by automated systems, compound challenges that emerge in economic aspects due to the high costs of implementation

Table 6

Key technologies and future opportunities for addressing challenges in adaptive and responsive HCI.

Challenge	Key technologies addressing challenge	Real-world applications	Future trends
Adaptivity	Behavioral biometrics (BBCA, BehaveSense)	Smartphone authentication, fraud detection	Continuous evolution of biometric methods
	Intelligent agent-based systems (CORMORANT, emotiondriven models)	Smart homes, personalized assistants	Enhanced AI driven authentication
	Edge computing for dynamic risk assessment	Autonomous vehicles, industrial automation	Multimodal authentication for diverse environments
Real-time responsiveness	Edge computing for latency reduction (LLRIS framework, MAR frameworks)	AR games, remote surgery	Integration of federated learning and reinforcement learning
	IoT in smart cities, healthcare	Traffic management, wearable health monitors	More energy efficient, real-time processing architectures
Cost and complexity	Prototyping, usability evaluation tools	Product design, user feedback loops	Low-cost HCI solutions with scalable designs
	Agent based mixed reality systems	Virtual museum tours, collaborative design	Simplified deployment frameworks
Privacy and security	Behavioral biometrics for continuous monitoring	Online banking, employee monitoring	Privacy preserving data handling techniques
	Quantum HCI	Cryptography, secure data centers	Postquantum secure HCI development
Immersive UX	Emotion recognition systems (RubikBiom)	Virtual therapy, emotional ai	Ethical and regulatory frameworks
	Virtual/AR	Virtual training, retail experiences	More immersive environments with reduced energy consumption
	Conversational interfaces	Customer service bots, virtual assistants	Emotional intelligence for conversational agents
Portability	Interactive visualizations	Financial dashboards, data journalism	Enhanced personalization in HCI systems
	Wearable sensors	Fitness trackers, remote patient monitoring	Expansion of wearables in healthcare and fitness
	Inertial measurement units	Motion capture in sports, physiotherapy	Lightweight, adaptive HCI systems
Energy efficiency and sustainability	Neuromorphic computing for wearables	Smartwatches, AR/VR glasses	AI-optimized low power HCI devices
	Smart grids with edge processing	Energy monitoring, home automation	Green computing principles in HCI design

against an uncertain Return on Investment. Thus, unlocking adaptive HCI's fullest potential into the manufacturing environment requires comprehensive strategies integrating technological developments, human resource improvement, open communication, and ethical consideration.

In conclusion, by systematically addressing technical, ethical, and inclusion-related challenges with tailored mitigation strategies, the proposed model evolves from a conceptual framework into a technically feasible, scalable, and adaptable solution for real-world industrial environments. These strategies ensure dynamic data processing, adaptive workflows, and real-time interactions remain efficient and secure while fostering trust, fairness, and operational excellence. By prioritizing user-centric design, inclusivity, and ethical considerations, the model paves the way for a secure and inclusive implementation, aligning with the demands of modern SM.

5.3. Future trends in AHCI

To effectively address the previously presented technologies and challenges, Table 6 summarizes the emerging trends and advancements in HCI to provide innovative solutions, enhancing the adaptability and resilience of the proposed framework.

This table outlines the evolving role of key technologies, like behavioral biometrics, intelligent agent-based systems, and edge computing, in addressing challenges in adaptivity, real-time responsiveness, cost, privacy, security, immersive UX, portability, and energy efficiency in HCI. The future trends emphasize the need for continuous innovation in AI (Shahzad and Qazi, 2020), ML, quantum computing, and energy-efficient solutions, enabling more personalized, secure, and sustainable HCI systems across various industries.

Moreover, advancements in AI and ML techniques can enhance the adaptive HCI framework. The operators' training and failing diagnosis are being revolutionized with immersive environments allowing for real-time interventions simulating complex scenarios. In such scenarios, the operators can experience and respond to the challenges as they happen. System decisions also get improved over time in terms of fault

detection, diagnosis, etc. They keep refining their decisions through reinforcement learning.

XAI could play an important role in addressing operator trust issues by providing transparency into decision-making processes. This way, greater acceptance and trust in the machine increases. In addition, Generative AI helps in building multilingual and contextually aware interfaces, which can facilitate seamless communication for diverse users.

Another key aspect is the integration of cognitive computing, emotion recognition, and physiological sensing logging technologies. Dynamic adaptation to cognitive states and stress levels of operators improves decision accuracy and reduces errors within high-performance environments.

6. Conclusion

This paper offers a comprehensive review of adaptive HCI and indicates the need for modern, continuous, and non-intrusive authentication mechanisms. It presents critical technologies, such as behavioral biometrics, agent-based intelligent HCI systems, and edge computing to achieve real-time adaptive and secure HCI. The study proposes to design HCI systems within FDD paradigms to dynamically modify themselves with respect to user behavior and environment context for secure and seamless interaction.

As behavior and context evolve over time, adaptability is very important, particularly on a more relaxed level. It will bring a difference to the everyday experience of employing systems from the standpoint of having an argument in favor of using security-aligned systems. Adaptable systems will deliver better performance in terms of UX while keeping higher standards associated with security and being truly usable for all, managing ever-increasing complexity in modern computing environments. Efficient adaptation algorithms will be part of HCI in future and protection against illegal access will remain high. It seeks to provide a special touch on the agenda of making I4.0-I5.0 space that focuses more on human-centricity. As I4.0 focused on how to maximize automation and efficiency, I5.0 shifts more to the human being involved,

making well-being and security of the operator the number one priority. By including adaptive, secure, and user-sensitive HCI systems, this direct approach invariably creates a seamless transition towards I5.0, where seamless and secure human-centered interaction is paramount. This ensures that technology enhances humans rather than compressing them.

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Chihi Inés: Writing – review & editing, Supervision, Funding acquisition. **Rania Hamdani:** Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization.

Author Agreement Statement

We declare that this manuscript is original, has not been published before, and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used was simulated by open source tools

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